

Economic Complexity HW2

Productivity and the Density of Economic Activity (Ciccone & Hall, 1996)

- Research Question: Do areas with a higher spatial density of economic activity have higher productivity?
- Data: The authors used U.S. data from 1988. They combined state-level data on Gross State Product (as a measure of output) with more granular, county-level data on employment and land area.
- Method: The authors used an econometric model to link state-level productivity (Gross State Product) to a specially constructed index of county-level employment density. This allowed them to measure the impact of density itself, separate from other factors.
- Findings: Density has a strong, positive, and causal effect on productivity.
 - The main finding: Doubling employment density in an area increases average labour productivity by 6%. This effect is substantial and can explain more than half of the observed variation in productivity across U.S. states.

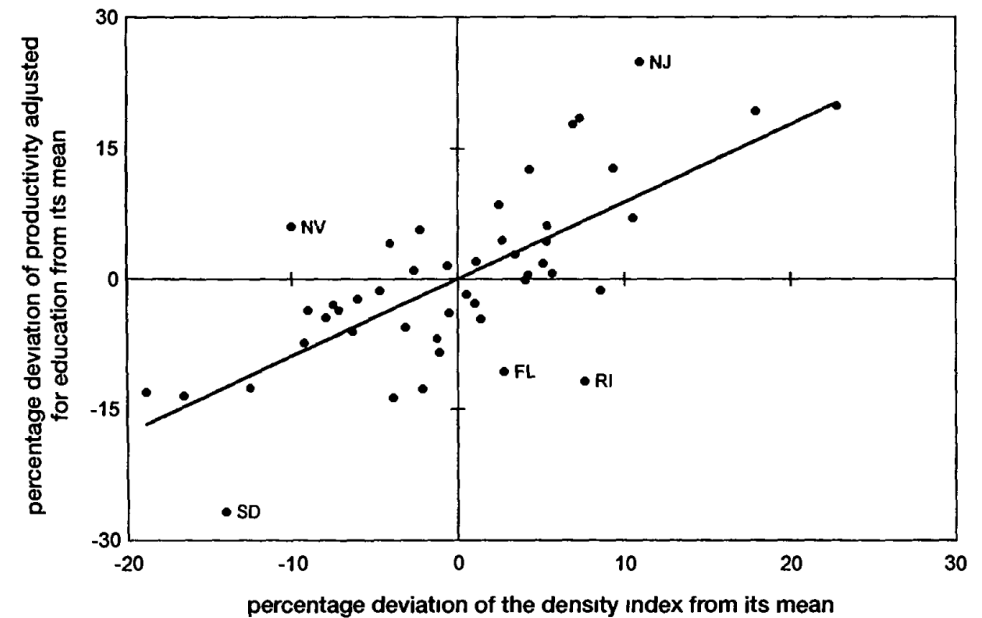


FIGURE 3. DENSITY AND PRODUCTIVITY ADJUSTED FOR EDUCATION, $\theta = 1.058$

Chetty et al. – Social Capital I: Measurement and Associations with Economic Mobility (Nature, Vol. 608 (2022))

Research question: Which form of social capital best predicts upward income mobility?

Data: 21B Facebook friendships (72M users, ages 25–44) – plus Opportunity Atlas, US Census, Civic data

Methods

- Mapped across U.S. counties and ZIP codes
- Regression analysis – EC, Cohesion, Civic Engagement
- Casual tests – Reverse causality, Selection ruled out (race-homogeneous areas), Quasi-experiment confirms causal EC → mobility effect
- Interpretation – Cross-class friendships

Findings

Key results

- Three distinct, weakly correlated dimensions: EC, Cohesion, Civic Engagement
- EC stable over time; varies geographically across U.S.
- EC strongly correlated with upward mobility ($r = 0.65$); others weak or none
- EC–mobility link robust at ZIP level ($r = 0.69$)

Causal evidence

- Reverse causality ruled out (childhood EC)
- Selection ruled out (race-homogeneous areas)
- Quasi-experimental moves confirm causal EC → mobility effect
- +1 EC → +9.8 income percentiles for low-SES children

Interpretation & policy

- EC mediates Cohesion and Civic Engagement effects
- Cross-class friendships drive opportunity
- Fostering cross-SES ties could close 37% of the opportunity gap

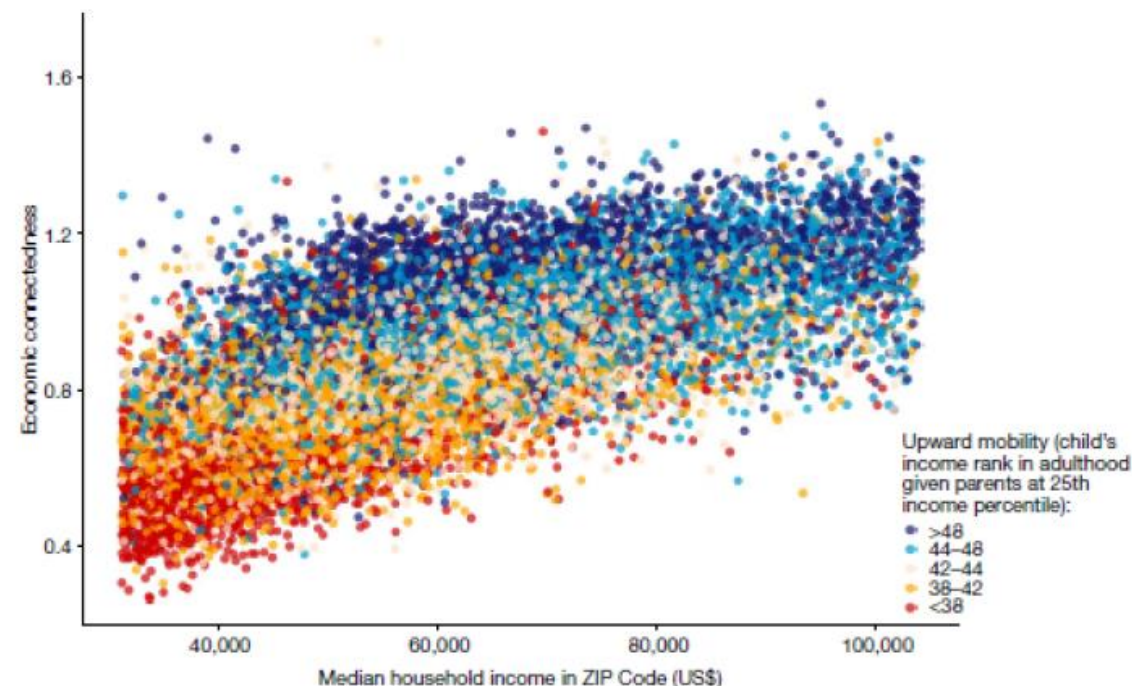


Fig. 6. Upward mobility rises with economic connectedness

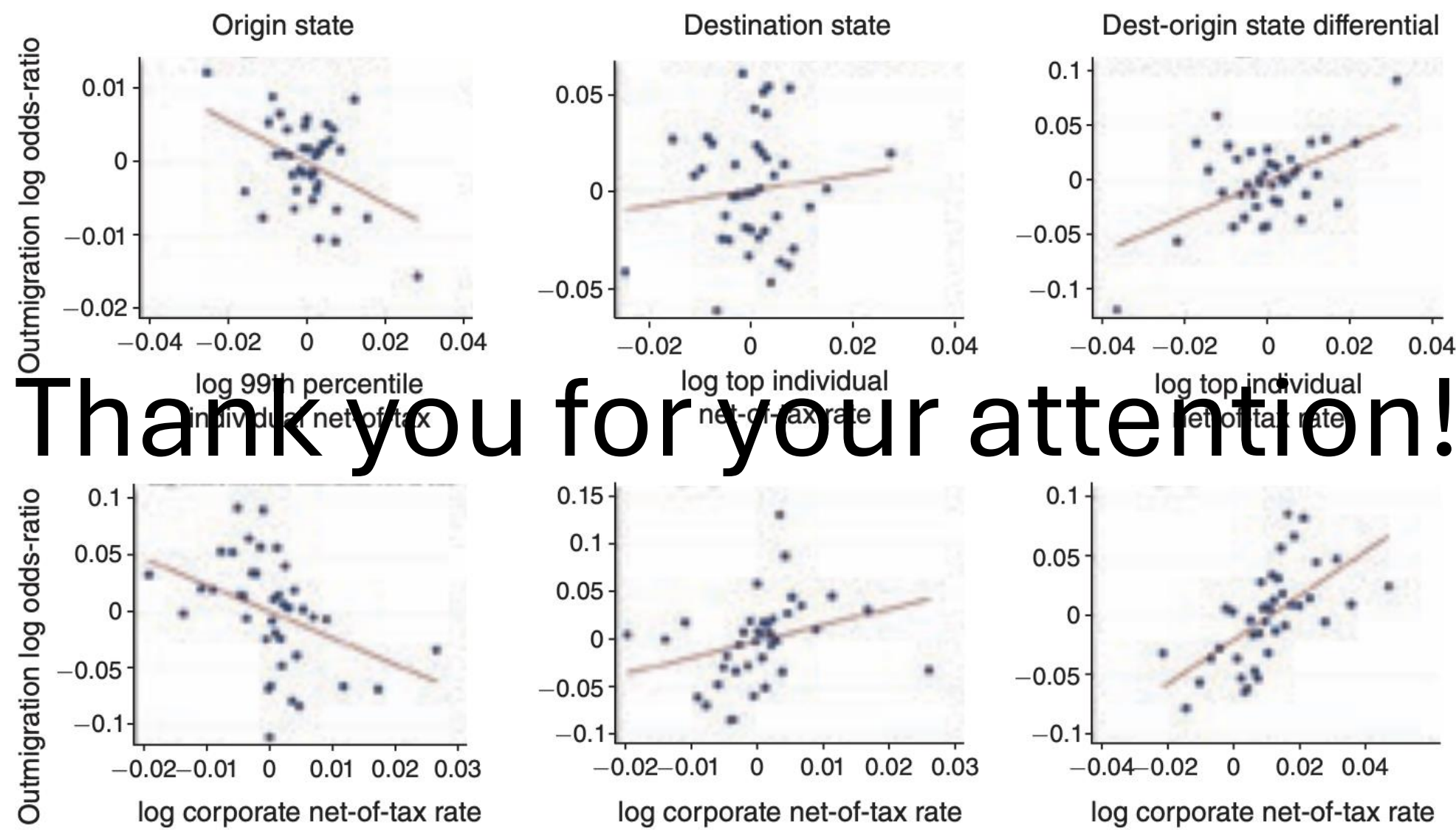
Figure: it illustrates the core finding of the study: upward income mobility increases with higher economic connectedness, showing that cross-class friendships — rather than local income levels — drive opportunity.

Defining urban clusters to detect agglomeration economies

Cottineau, C., Finance, O., Hatna, E., Arcaute, E., & Batty, M. (2016). Defining urban clusters to detect agglomeration economies. *Environment and Planning B: Urban Analytics and City Science*, 46, 1611 - 1626.

- **Research question:** Are agglomeration economies specifically urban? Does the location of economic agents within cities matter? Are larger cities more unequal?
- **Data:** France — 2011 census communes (36546), 2011 tax office household incomes brackets, 2008 CLAP firm-wage data.
- **Method:** log-log scaling regressions, and evaluate across the D-C-P parameter space.
- **Findings:** Wage superlinearity is strongest in dense job centers; population-based effects appear regionally. High incomes cluster in dense cores, and larger cities show a slight but consistent rise in inequality.





Thank you for your attention!

FIGURE 4. OUTMIGRATION VERSUS NET-OF-TAX RATES

The Role of Spatial Effects in Regional Concentration

(Giuseppe Arbia, 2001)

- Key Issue: Traditional concentration measures (e.g., Gini) ignore geography.
- Core Proposition: Regional concentration is not only explained by a-spatial variability, but spatial clustering as well, where:
 - A-spatial measures focus on concentration, irrespective of location.
 - Spatial measures capture neighbor effects and polarization.
- Methodology: Italian manufacturing employment by province (1991) analyzed using:
 - Gini index (a-spatial)
 - Moran's I (spatial autocorrelation)
 - Getis-Ord index (local clustering)
- Findings: Measures are weakly correlated → each reveals different patterns.
- Conclusion: Combine spatial and a-spatial metrics for full understanding of concentration.

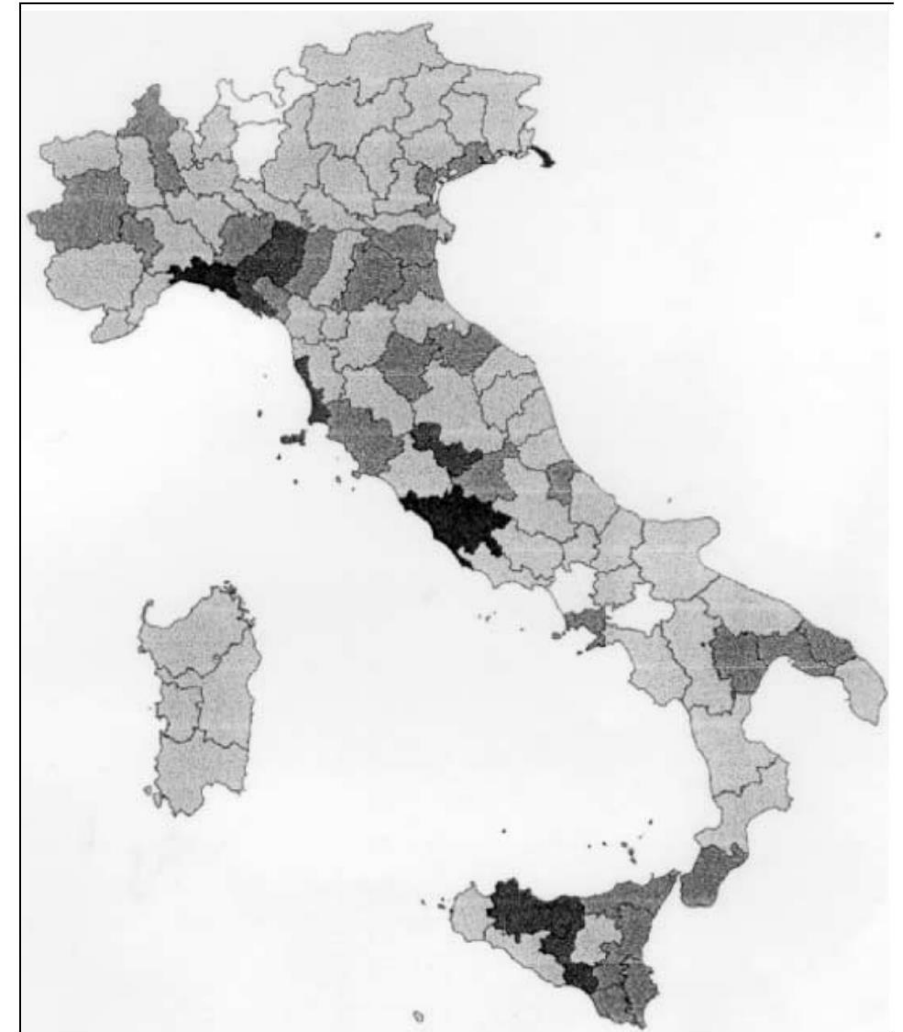


Fig. 5. Map of concentration of manufacturing employment in Italy in 1991. Grey-scale colours refer to the Gini location index. There are 5 classes: class 1 (*white*) refers to low concentration with G ranging from 0 to 0.1; class 2 refers to G ranging from 0.1 to 0.4; class 3 refers to G ranging from 0.4 to 0.6; class 4 refers to G ranging from 0.6 to 0.8 and class 5 (*black*) refers to G ranging from 0.8 to 1

COVID-19 Spread in Beijing

Research paper: [Spatial distribution characteristics of the COVID-19 pandemic in Beijing and its relationship with environmental factors – Yi Han et al. 2021](#)

Research question:

- How did COVID-19 spread in Beijing, and what environmental factors affected this spread?

Data Used:

- COVID-19 case numbers in Beijing from June 11 to July 5, 2020.
- Environmental factors like population density, distance to Xinfadi market, and other city features.

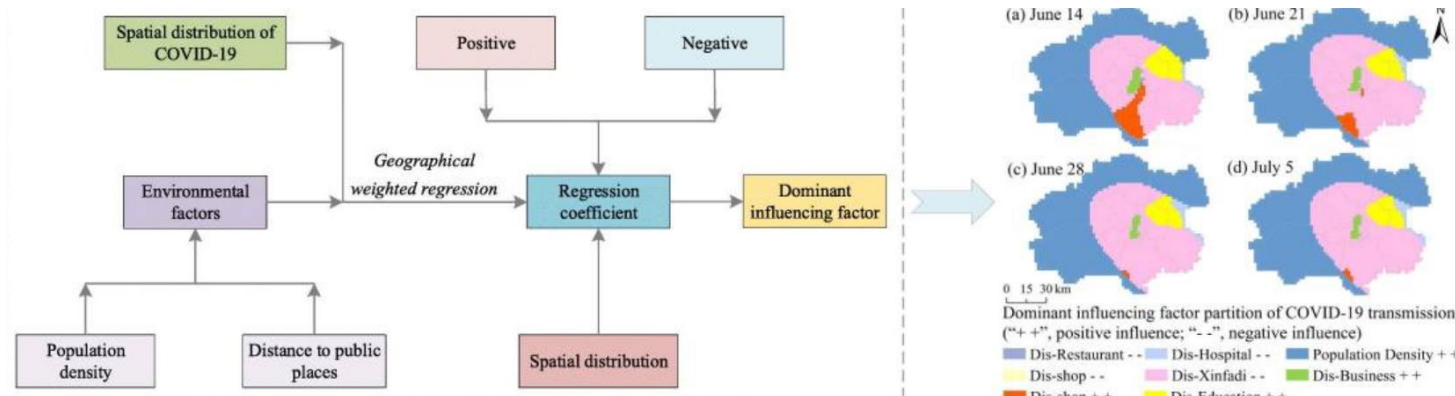
Methods:

- Checked for clusters using spatial autocorrelation.
- Used Spearman correlation to see which factors were related to case numbers.
- Applied Geographically Weighted Regression (GWR) to see how factors influenced different areas.

Findings:

- COVID-19 spread in two main phases: first spreading outward, then concentrating near existing cases.
- Areas around the Xinfadi market had higher case clusters, showing it was a hotspot.
- Population density and closeness to the market were the strongest factors affecting the spread.

Graphical abstract:



IS INNOVATION (INCREASINGLY) CONCENTRATED IN LARGE CITIES? AN INTERNATIONAL COMPARISON

MICHAEL FRITSCH, MICHAEL WYRWICH

Is innovation becoming increasingly concentrated in large cities over time across different countries?

Data

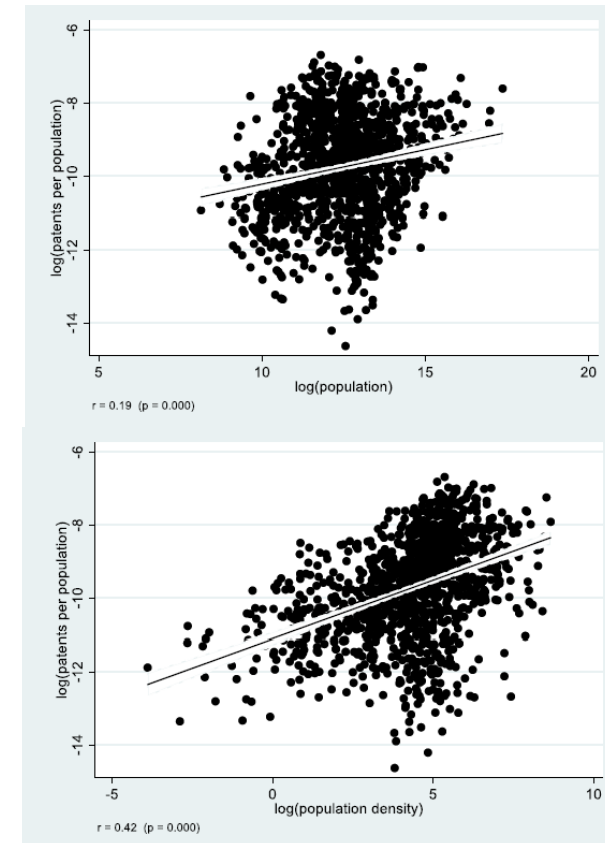
- OECD regional patent database (RegPat) covering 14 developed countries (2000-2015)
- Regional categories defined by the OECD functional urban areas

Methodology

- Compared patent concentration and innovation intensity: spatial distribution, patent-per-population ratios, and concentration indices (Herfindahl-Hirsch Index - HHI)
- Tested urban scaling and inventor productivity by regressing patent counts on regional population and inventor numbers, including checks for patent quality (citations, renewals, originality)

Findings

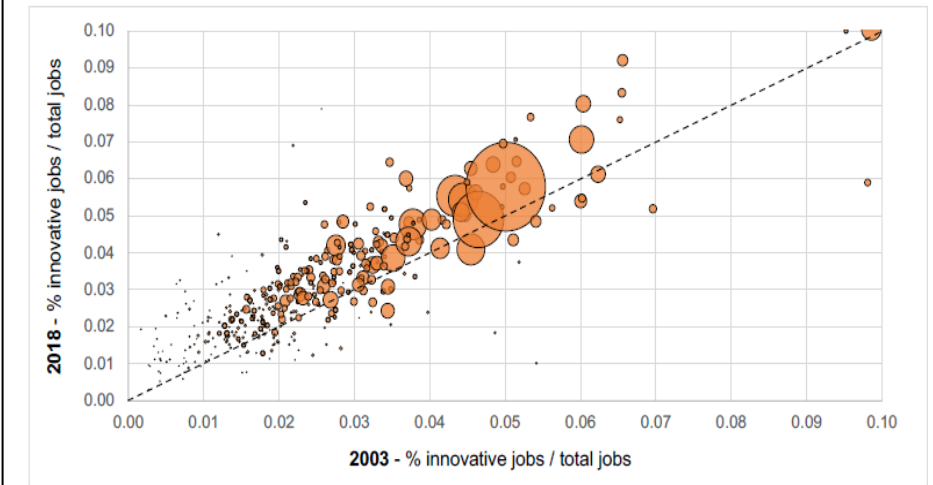
- The role of agglomeration economies is much less pronounced than many authors suggest
- Innovation is not increasingly concentrated in large cities, patents are more geographically concentrated than population but show no upward trend over time
- While big cities show higher patent counts, inventors are not more productive → innovation does not require large cities
- Centralizing R&D in metropolitan areas is inefficient, supporting regional and rural innovation is more balanced and effective



Milene Simone Tessarin & Carlos Roberto Azzoni (2023), Innovation and Competitiveness: The Regional Dimension

- Research question and focus
 - How does labour pools and geographic concentration influence regional innovation and the pace of technological change
 - Developing economies → human capital and workers skills → innovative potential outside R&D
- Methodology & Data
 - Unit of analysis: 374 labour market areas (LMA) across Brasil
 - Data: from RAIS → from 2003 to 2018 50 million formal workers
 - Measuring innovative potential by occupational skills → Six innovative skills via PCA: creativity, active learning, design, engineering/technology, technology design, updating knowledge → Innovation potential index = weighted sum of innovative occupations
 - Empirical model panel regression 2003-2018 4 non-overlapping 4 year periods → to test for convergence or divergence across regions
- Results
 - High concentration: 44 LMAs (12%) hold more then 80% of innovative jobs → close to 20-80 rule
 - Lagging regions grew faster → slow convergence (evidence from the regression as well)
 - Riceher regions are more innovative, industrial structure drives innovation, large firms are more likely to innovate → Schumpeter scale effect
- Conclusion
 - Regional path dependence matters, BUT redistributive policies helped the peripheries to catch up, still a big difference
 - Modest deconcentration of innovative potential unlike Europe
 - Still remains geographically concentrated

Figure 5 – Change in the share of innovative jobs by regions – Selected group (2003-2018)

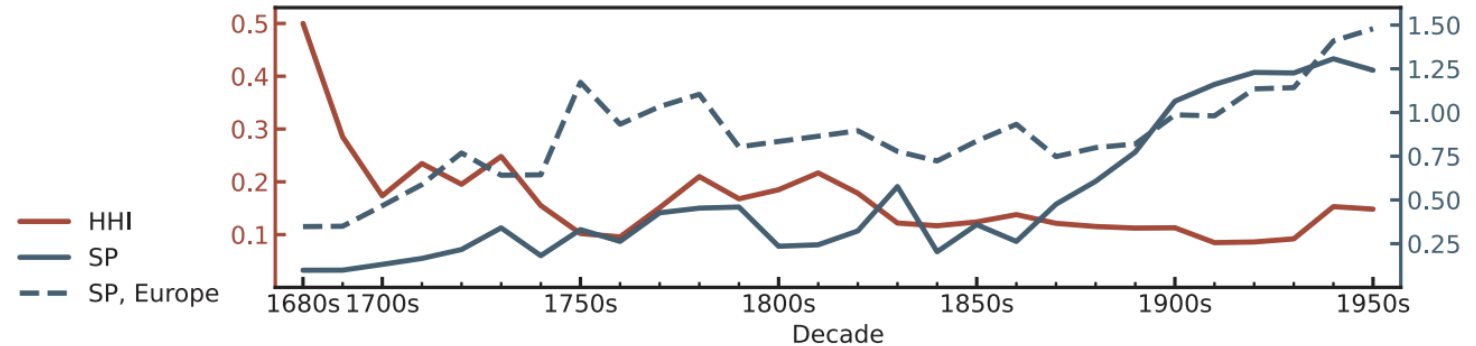


Source: Author's elaboration. Note: The bubble size represents the number of innovative jobs by region in 2003. Includes only regions with up to 10% of innovative jobs.

Quantification of the spatial-temporal patterns of great ideas (Li et al., 2023)

- *What are the spatial and temporal patterns behind the birth of great ideas in human history?*
- *Are they geographically concentrated, and how have those centers shifted over time?*

- **2001** books/papers by **1168** notable authors
- **9 disciplines** (math, physics, biology, etc.)
from **129 cities** (1687–1953)



- **Spatial-temporal bipartite network** linking *cities* & *decades*
- Used **Herfindahl-Hirschman Index (HHI)** and **Spatial Separation Index (SP)** to measure concentration
- Developed a **decade-space network** to detect periods of structural change
- Created new metrics: **City Leadership (Lc)** and **Decade Prosperity (Pd)** via iterative network algorithm
- **Great ideas are more geographically concentrated** than other human activities
- **Top 5 cities (London, Paris, New York, Leipzig, Berlin)** produced **more than 50%** of all works
- Identified a “**Great Transformation**” around the **1870s**, marking a global academic shift from **Europe to the United States**
- The **spatial pattern** of idea generation mirrors **economic and social transitions** (industrialization, globalization)

Optimized data center site selection – mesoclimatic effects on data center energy consumption and costs.

Energy Efficiency **14**, 33 (2021).

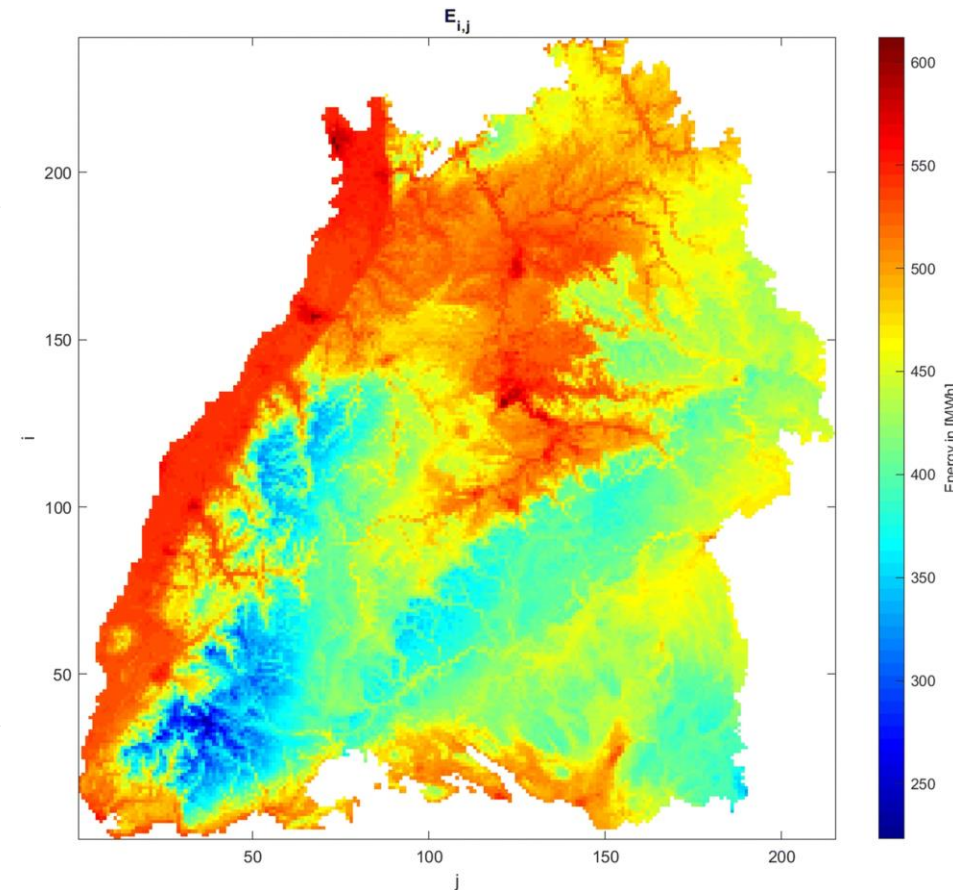
<https://doi.org/10.1007/s12053-021-09947-y>

This paper explores, how climate influences a data center's energy demand, thus revealing one key factor that influences where data centers will cluster regionally. Climate matters, due to the cooling need of data centers, where they can receive 'Free cooling' from the environment.

Data: The paper used energy consumption and heat data of data centers and aggregated temperature data from the German state of Baden-Württemberg.

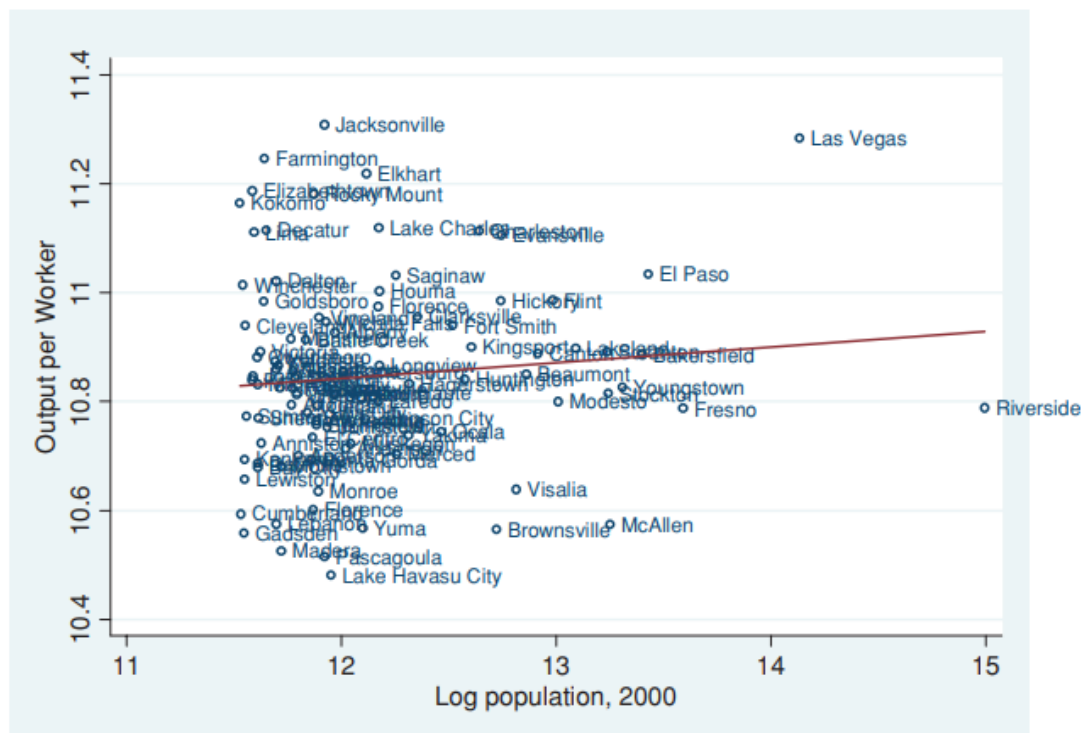
Method: They created models for calculating energy consumption by location and models of how much energy can data centers save by relocating 10, 25, 100 kms.

Findings: With on average reducing energy consumption 9.12% when relocating in a 10 km range, and 37.35% relocating in 100 km. Climate plays a crucial role in the decision of a data center's location.



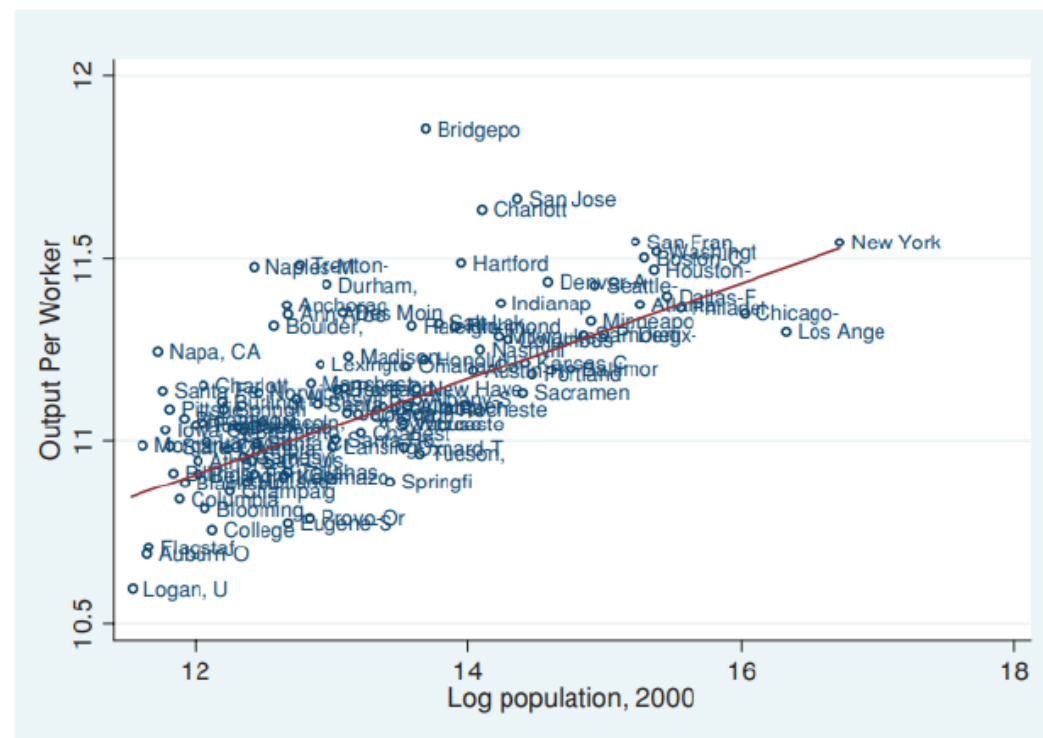
The Complementarity between Cities and Skills

By Edward Glaeser



Notes: (i) Units of observation are MSAs under the 2006 definitions with populations above 100,000 and where the share of adults with college degrees is less than 17.65 percent. Labor force and population is from the Census, as described in the Appendix. Gross Metropolitan Product is from the Bureau of Economic Analysis.

(ii) The regression line is $\text{Log GMP per capita} = 0.028 [0.028] * \text{Log population} + 10.50 [0.34]$. $R^2 = 0.01$ and $N = 100$.

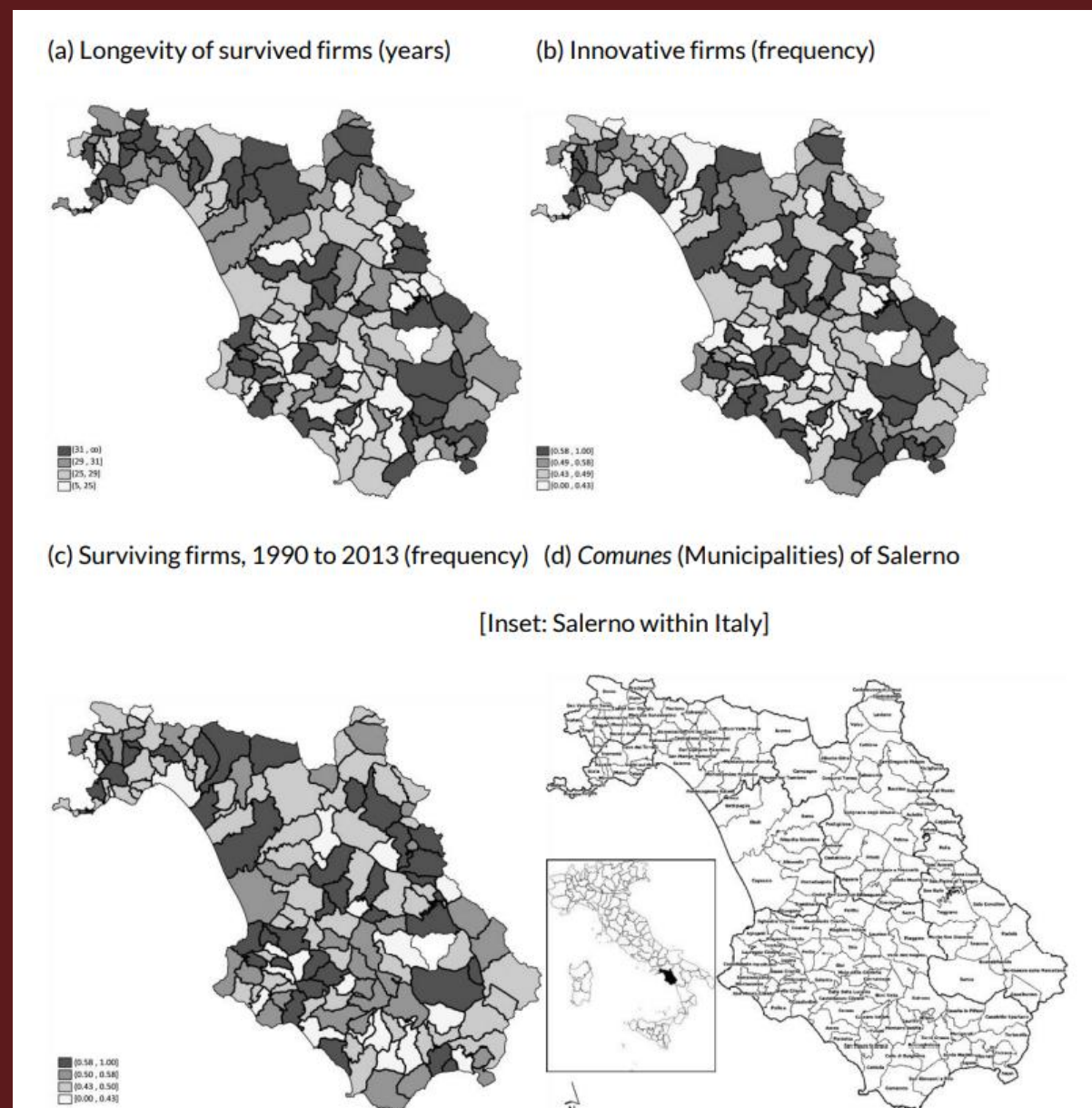


Notes: (i) Units of observation are MSAs under the 2006 definitions with populations above 100,000 and where the share of adults with college degrees is greater than 25.025 percent. Labor force and population is from the Census, as described in the Appendix. Gross Metropolitan Product is from the Bureau of Economic Analysis.

(ii) The regression line is $\text{Log GMP per capita} = 0.128 [0.015] * \text{Log population} + 9.46 [0.19]$. $R^2 = 0.44$ and $N = 100$.

SPATIAL AGGLOMERATION, INNOVATION AND FIRM SURVIVAL FOR ITALIAN MANUFACTURING FIRMS (BHATTACHARJEE, MAIETTA & MAZZOTTA, 2023)

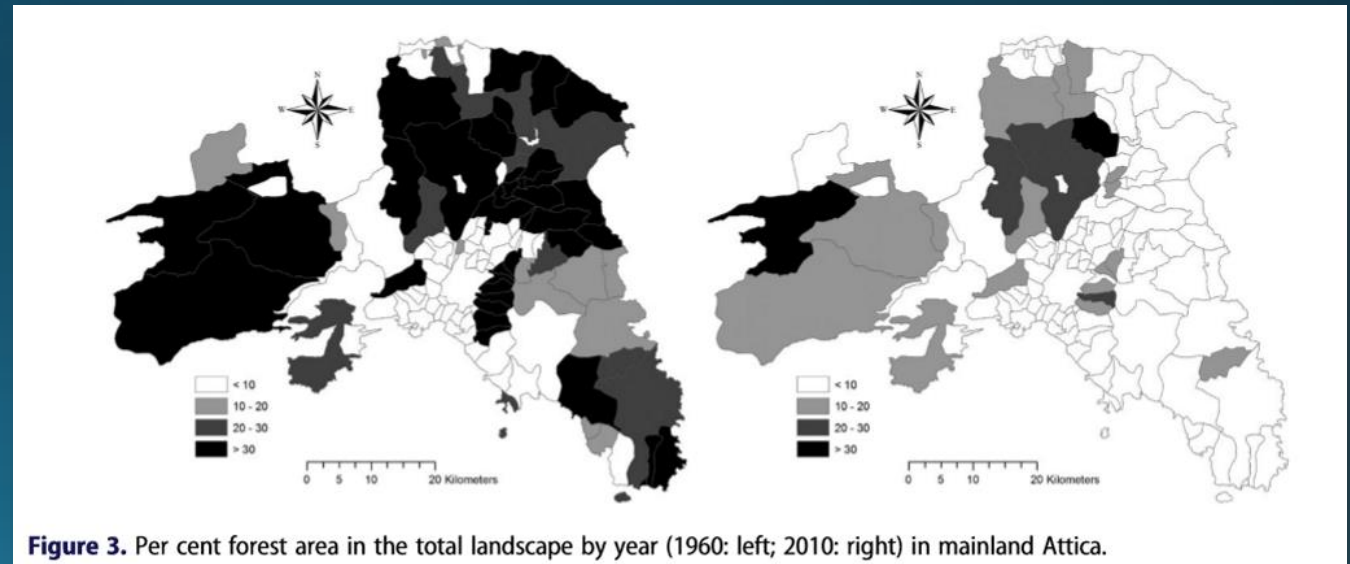
- 10 Research Question: How does spatial agglomeration affect firm innovation and survival in Italian manufacturing
- 10 Data: Firm-level panel data from southern Italy
- 10 Spatial econometric (Spatial Durbin) models to measure local spill-over effects
- 10 Findings: Firms near other innovative firms are more likely to innovate and survive
- 10 Conclusion: Local innovation clusters strengthen regional competitiveness and firm resilience



Back to Von Thünen: A Southern European Perspective on Mono-centric Urban Growth

(Colantoni et al., 2017 – International Planning Studies)

- How does monocentric urban growth affect land use and natural areas in Southern Europe?
- Data:
 - Attica region, 1960-2010
 - 26 socioeconomic, environmental & land-use indicators
- Methods:
 - Statistical analysis
 - Correlation
 - Multiway factor analysis
- Key findings:
 - Urban expansion reduces forest and cropland near the core.
 - Planning tools can limit land loss

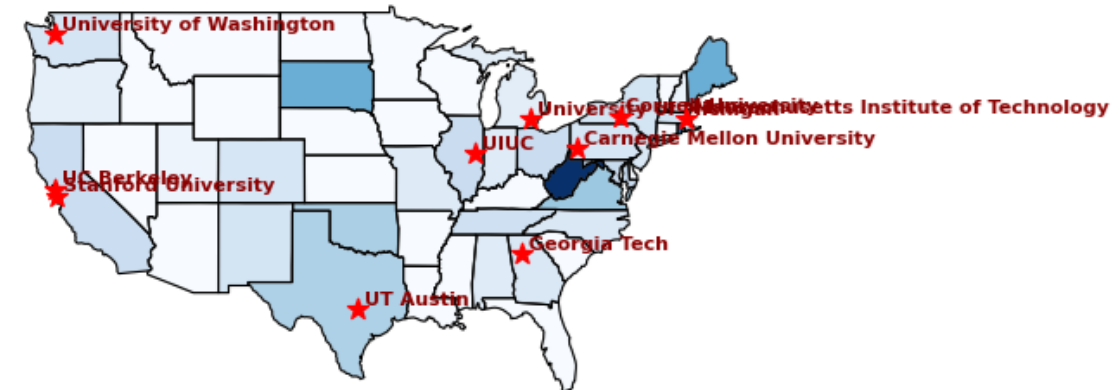


The spatial distribution of public support for AI research

The spatial distribution of public support for AI research By: Farhat Chowdhury, Albert N. Link, Martijn van Hasselt Chowdhury, F., Link, A. N., & van Hasselt, M. (2022). The spatial distribution of public support for ai research. Science and Public Policy, 49(4), 573–579. <https://doi.org/10.1093/scipol/scac008>

Spatial Distribution of AI-Focused SBIR Phase II Projects (FY 2020) with Top 10 AI Universities

- Between US firm awarded for AI research
- Top 10 universities ranked by their AI research importance
- High correlation in states
- Access to AI research resources are fundamental
- AI research fundings are geographically concentrated
- Policy makers should consider to wider the spatial distribution of funds
- <https://colab.research.google.com/drive/1kwOFSiDr8MQoEqF48oBhB9gzDSLILSPK?usp=sharing>



Balland et al: Complex Economic Activities Concentrate in Large Cities (2020)

Research Questions:

Why do certain economic activities (e.g. technologies, industries, scientific fields) strongly concentrate in a few large cities?

What role does the complexity of an activity play in this spatial concentration? Has this concentration changed over time?

Data:

353 US Metropolitan Statistical Areas (MSAs)

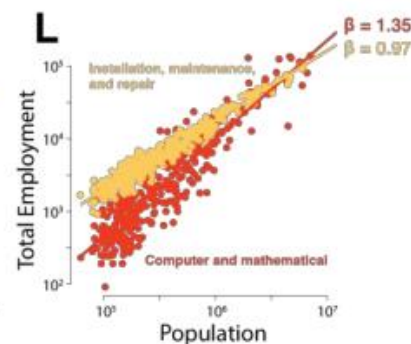
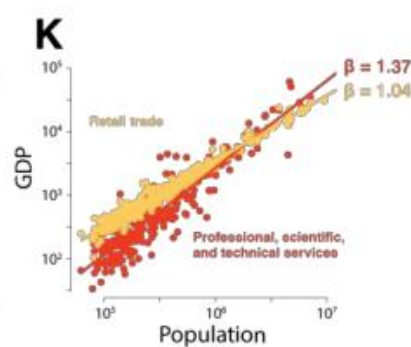
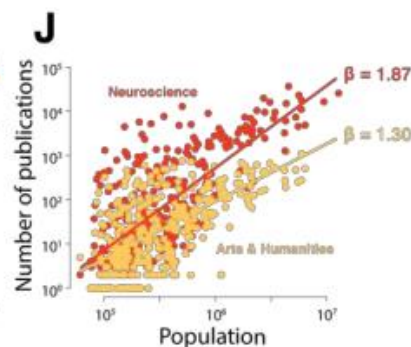
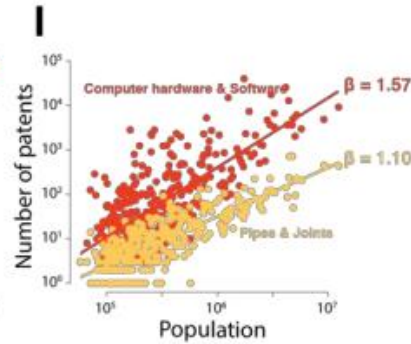
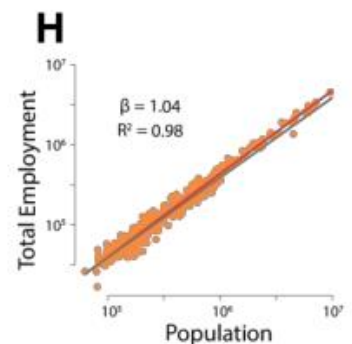
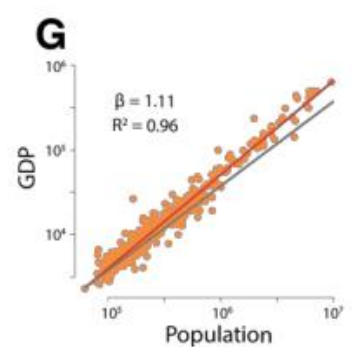
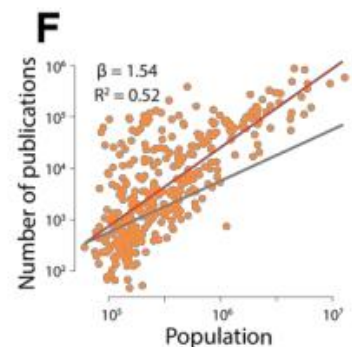
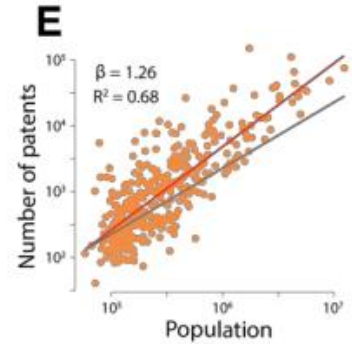
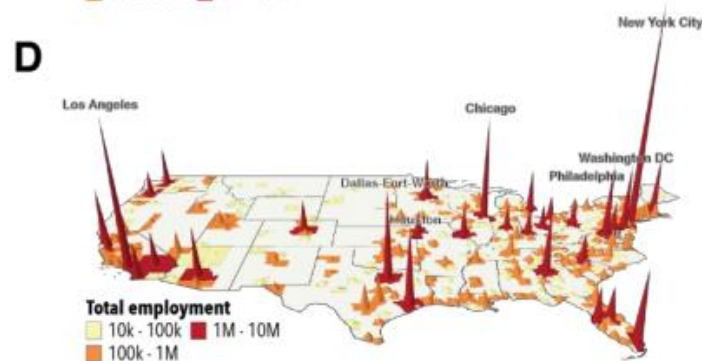
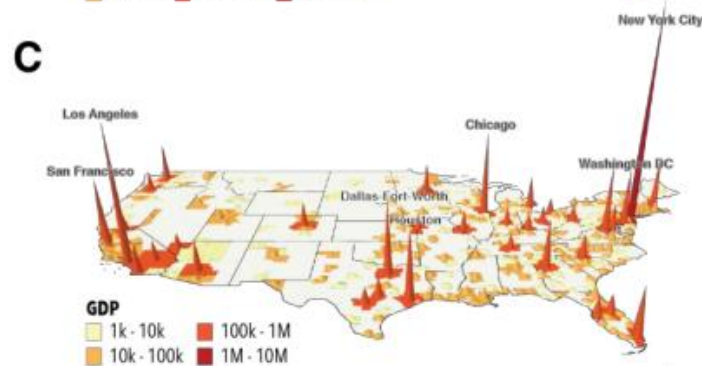
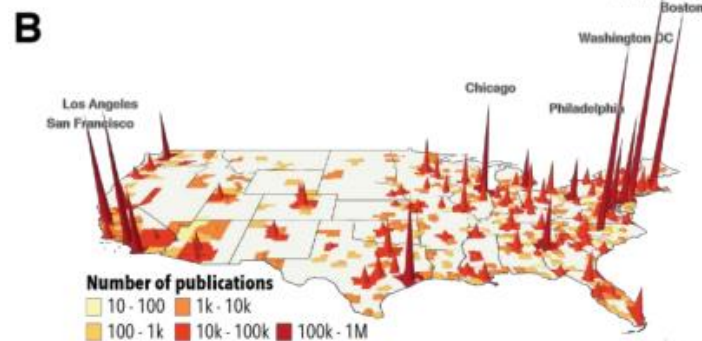
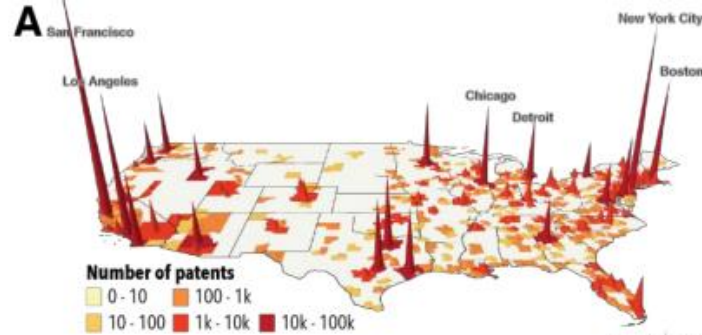
Indicators:

- Patents
- Scientific publications
- Industrial output
- Employment

Historical data (esp. patents) from ~1850 to present

Methods:

- Urban scaling analysis: Examines how activity levels scale with city size (superlinear, linear, sublinear relationships)
- Complexity measures: Based on collaboration size, technology novelty, knowledge combinations, etc.
- Time-series analysis: How concentration trends have evolved across decades for different complexity levels



Key Findings:

1. More complex activities are more spatially concentrated

- Activities with high complexity (e.g. biotech, neuroscience) are disproportionately located in a few major cities
- Complexity explains 40–80% of variance in spatial concentration (across industries, technologies, occupations, scientific fields)

2. Concentration increases over time

- The most complex technologies have become increasingly concentrated in the largest cities since the 1850s
- Even in the digital age, physical proximity remains critical for complex production & innovation

3. Reinforcing feedback loop

- Big cities attract complex activities → this fosters further urban agglomeration
- Smaller cities struggle to attract or retain such activities, leading to divergence

Takeaways:

- Innovation and complex production are not spreading evenly — they cluster in the largest cities
- Urban policy & regional development must consider the role of complexity and relatedness when promoting industrial growth
- Complexity is a key driver of spatial inequality in modern economies



THIRD PLACES AND NEIGHBORHOOD ENTREPRENEURSHIP: EVIDENCE FROM STARBUCKS CAFÉS

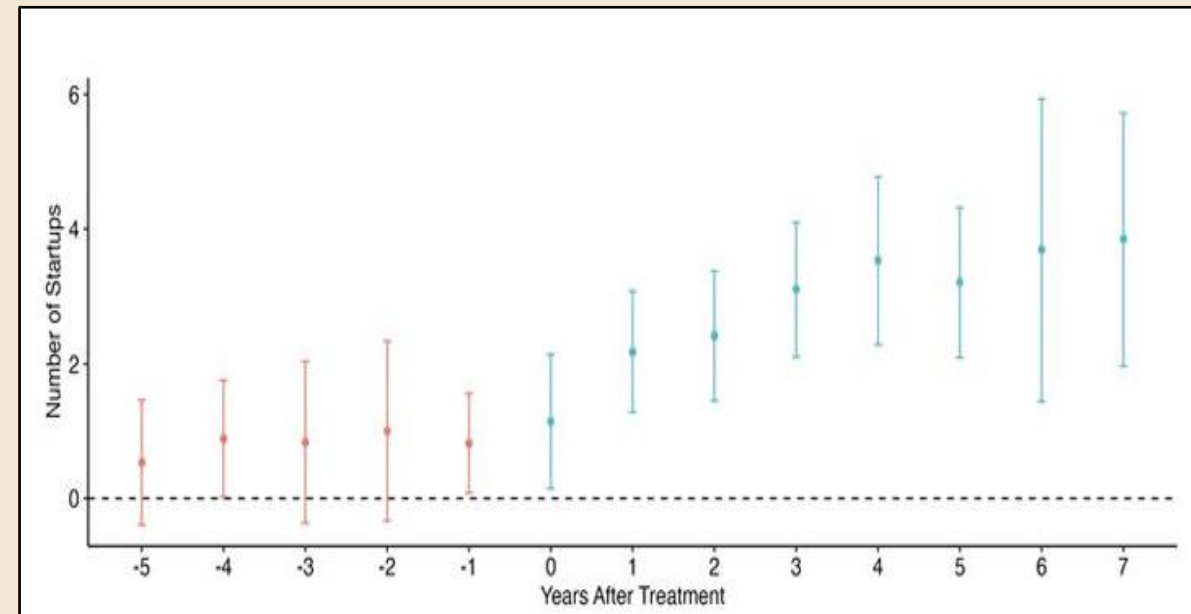
Jinkyong Choi, Jorge Guzman & Mario L. Small
(2024)

Presented by:
Réka Antal

Research Summary



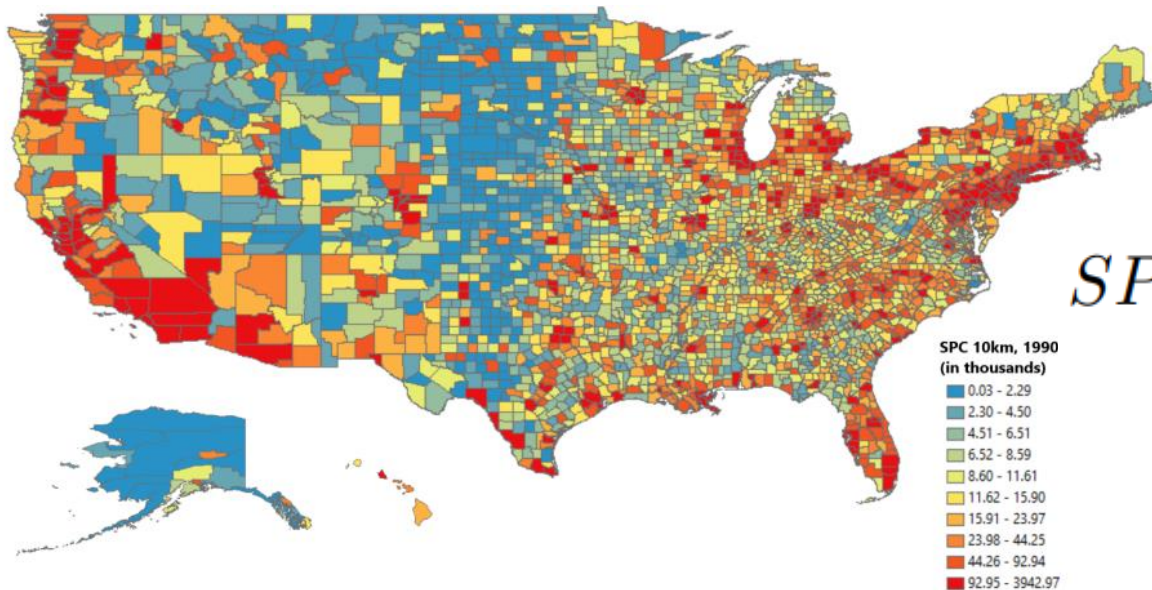
- **Research Question:** Do third places - especially cafés like Starbucks - boost **local entrepreneurial activity** by fostering social interaction and idea exchange?
- **Data:**
 - **main X** = 25,000+ Starbucks openings across the U.S. (2005–2021)
 - **Y** = new business registrations at the ZIP-code level
- **Methodology:** **Difference-in-differences** method comparing neighborhoods before and after Starbucks entry
- **Findings:**
 - **+4–6% increase** in new business registrations within 2 years of a Starbucks opening
 - Effects are strongest for **knowledge-intensive** and **creative** sectors (tech, design, professional services)
 - Mechanism: cafés act as **“third places”** - informal social hubs where entrepreneurs meet, share info, and build trust



The Impact of the Spatial Population Distribution on Economic Growth: Evidence from the United States

Constantin Bürgi, Nisan Gorgulu (2022)

- ▶ Basic idea: Concentration means less expense on high skilled workers + on product distribution
 - ▶ → Worth to implement new ideas → Population concentration correlates with GDP growth
- ▶ New measurement: Spatial Population Concentration (SPC) → Weighted average population surrounding every person within a geographic area
 - ▶ Better than population density and urbanization
- ▶ USA data from 1975 to 2015. Panel model regression with control variables and fixed effects
- ▶ Increasing the SPC by 1 standard deviation (in 25km radius), ceteris paribus GDP growth will be greater by 7.63%



$$SPC_d = \frac{\sum_{i=1}^N pop_i * n_{di}}{\sum_{i=1}^N pop_i} - 1$$

DOES RENEWABLE ENERGY TECHNOLOGICAL INNOVATION CONTROL CHINA'S AIR POLLUTION?

Yongfeng Zhu, Zhihong Wang, Xuefeng Wang, Lingling Zhu

A SPATIAL ANALYSIS

Research question:

What relationship can we observe between Renewable Energy Technological Innovation (RETI) and Air Pollution from a spatial perspective?

Dependent Variable:

- Air Pollution (proxied by SO₂ NO_x and PM₁₀ concentrations)

Explanatory Variable:

- RETI (proxied by number of renewable energy patents)

Control Variables:

- Population, GDP, Industry Structure, Environmental Regulation

Data:

- Gathered from China Environmental Yearbook and China Statistical Yearbook
- Balanced panel data
 - 31 Chinese provinces
 - Years between 2011 and 2017

Methods employed:

- Global and local Moran's I indices
- Simple and Spatial Durbin Panel Models

FIGURES, EQUATION, RESULTS, AND POLICY IMPLICATIONS

Figure 1.

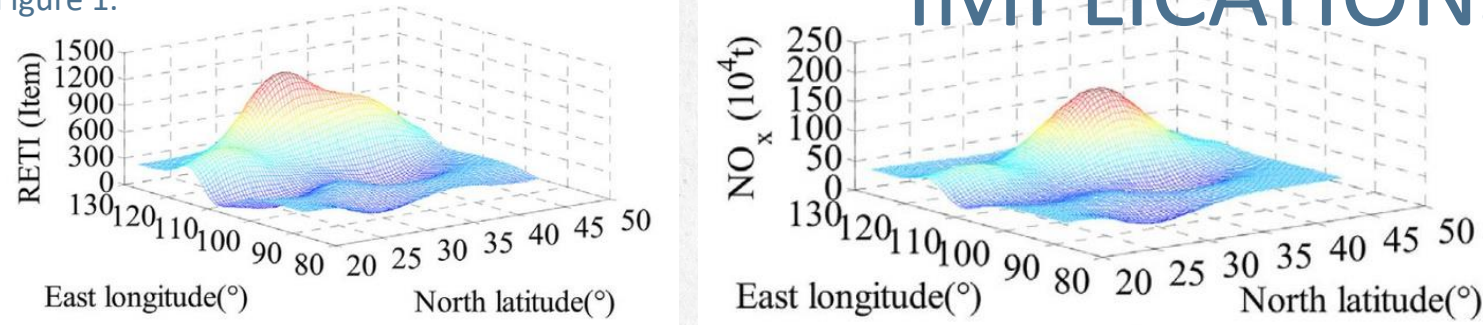


Figure 1. Spatial patterns of RETI and Air Pollution

$$\ln Y_{jt} = \lambda_0 + \lambda_1 \ln RETI_{it} + \lambda_2 \ln POP_{it} + \lambda_3 \ln GDP_{it} + \rho \sum_{j=1}^n W_{ij} \ln Y_{jt}$$

$$\delta_1 \sum_{j=1}^n W_{ij} RETI_{jt} + \delta_2 \sum_{j=1}^n W_{ij} POP_{jt} + \delta_3 \sum_{j=1}^n W_{ij} GDP_{jt} + u_i + v_t + e_{it}$$

Equation 1. Spatial Durbin Model Equation

Results:

- Air Pollution is spatially autocorrelated
- RETI has spatial spillover effects on Air Pollution
- RETI is negatively associated with Air Pollution

Figure 2.

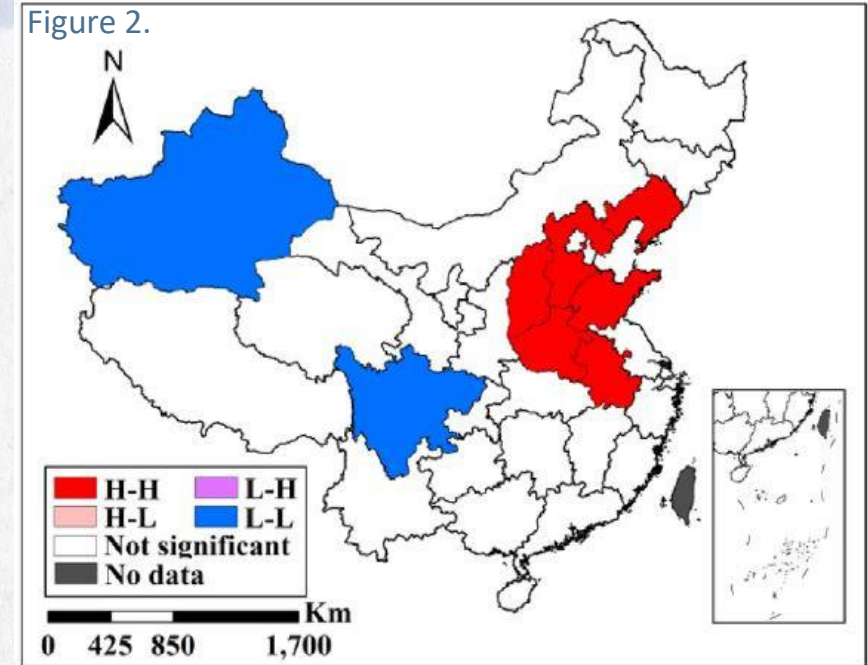
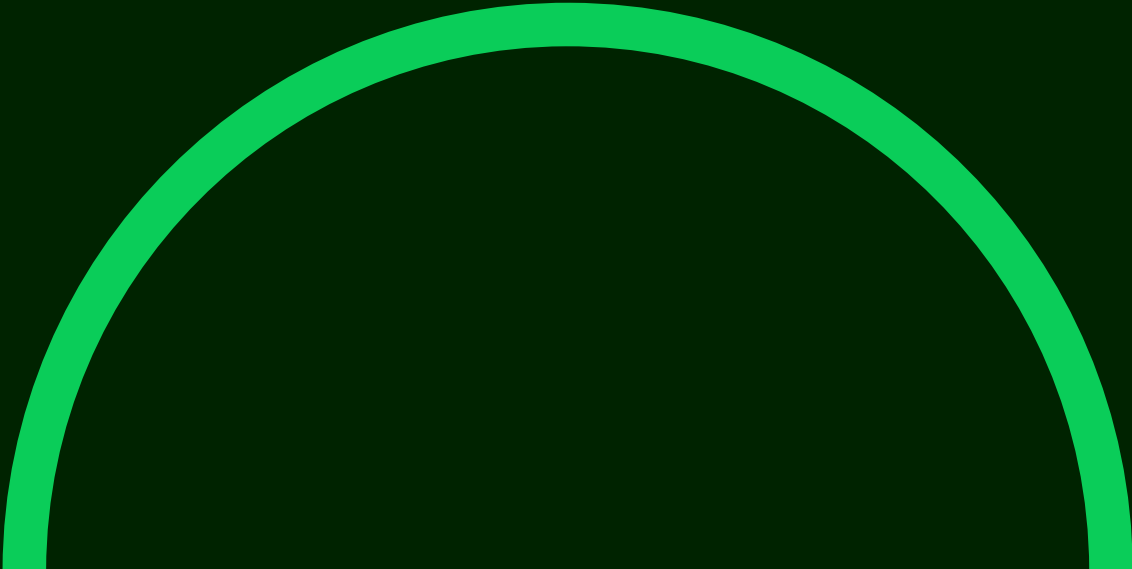


Figure 2. Spatial autocorrelations of RETI

Policy recommendations:

- Support of RETI in the North China Plain
- Establishment of a specialized community to share R&D talent, funds, and resources
- Promotion of industrial upgrade and the adoption of renewable energy sources



Robots and Jobs: Evidence from US Labor Markets

Acemoglu and Restrepo

Made by: Kristóf Andrási



Research: how did the rise of industrial robots between 1990 and 2007 affect jobs and wages in US local labor markets?

Data: USA and EU robot data to calculate how exposed each local area was to robots in the US
employment and wages + controls for other shocks like Chinese and Mexican imports, routine jobs and offshoring

Theory: Robots can replace people on some tasks (displacement effect) but they can also lower costs and prices -> expand output (a productivity effect) - the net effect can go either way

Method: they regress differences, weighted by population, fixed-effects

Employment: exposure to robots has a strong negative association with employment-to-population. In stacked-differences with commuting-zone fixed effects, the employment effect remains negative

Wages: The wage effect is negative and significant

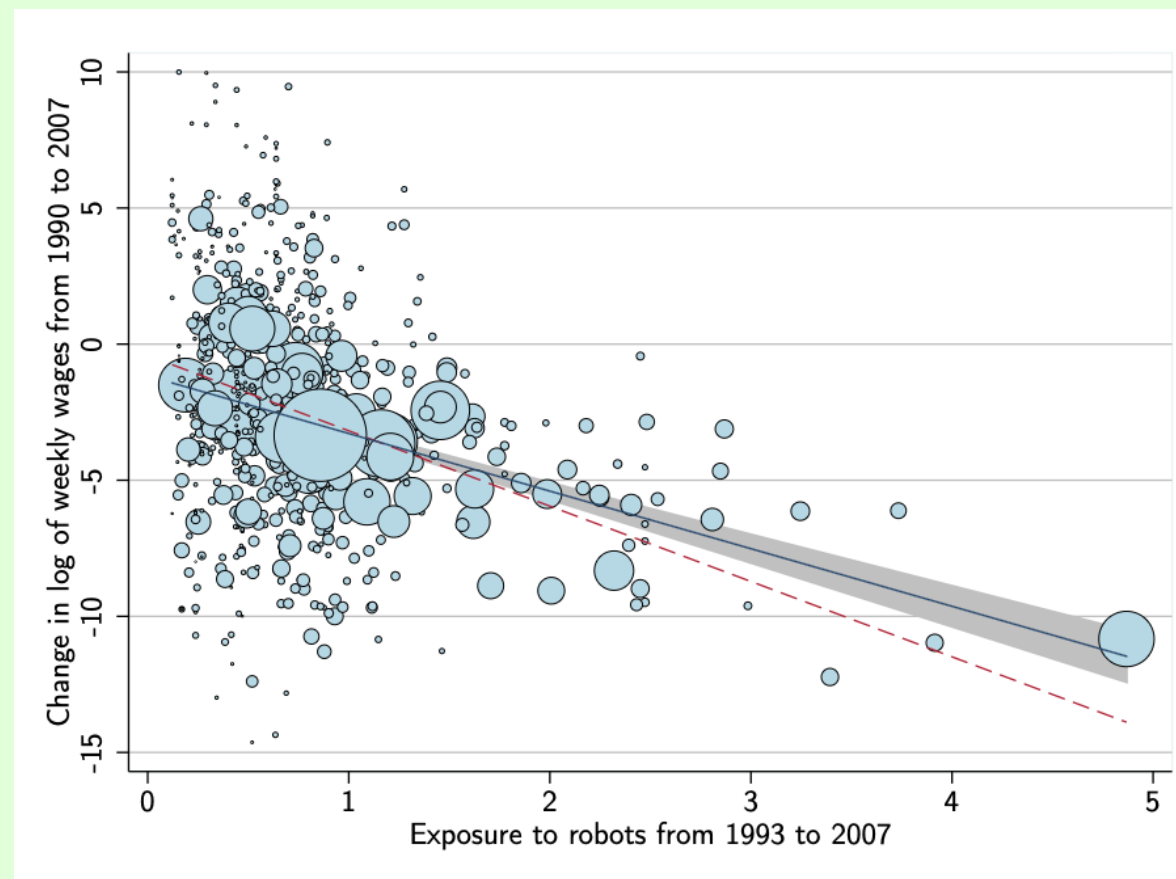
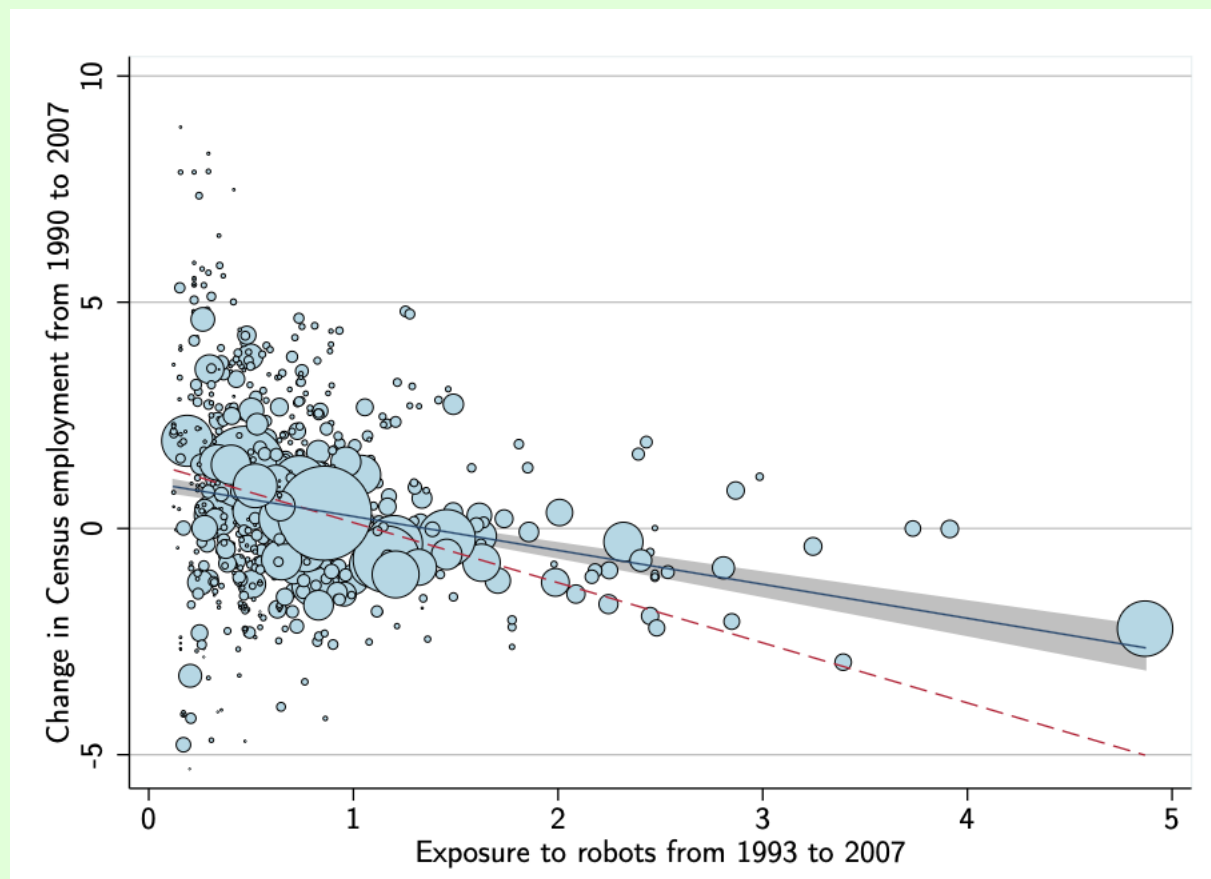
Gender: Effects are negative for both men and women, with men's employment hit about 1.5-2x more (wage effects are similar for both)

Job losses are largest in manufacturing, especially heavily robotized sectors (autos, electronics, metals) and in routine manual roles (operators, assembly)

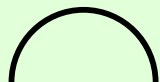
Education: For people without a bachelor's degree, jobs and salary went down, people above that education there was no clear effect

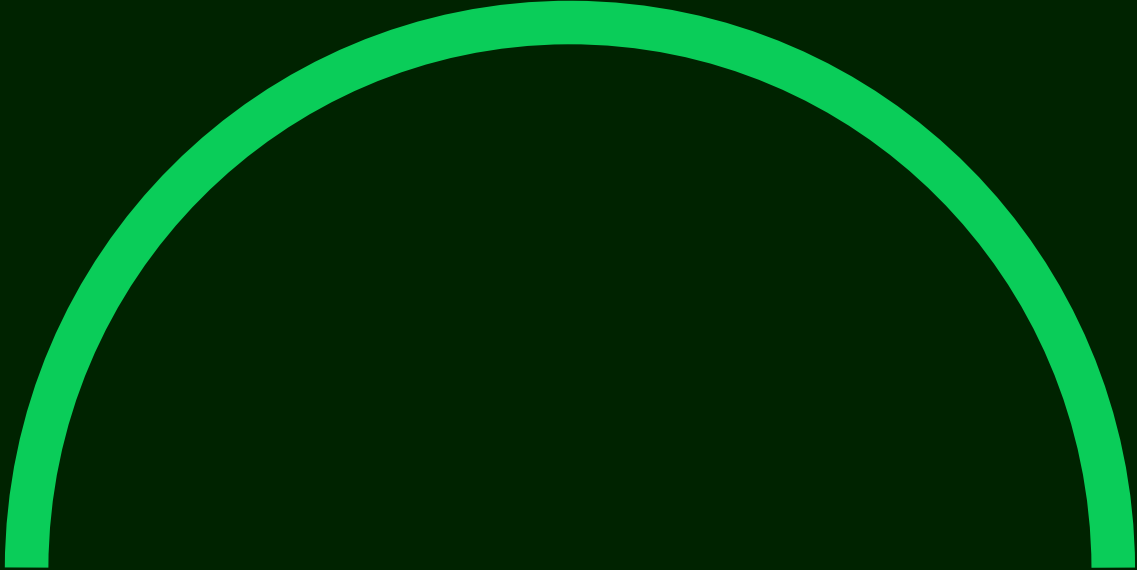
Robustness: Adding the control variables the robot effect is nearly unchanged

Employment and wages vs exposure to robots



source: *Robots and Jobs: Evidence from US Labor Markets* by Acemoglu and Restrepo





Thank you for your attention!

source: Acemoglu, D., & Restrepo, P. (2017). Robots and Jobs: Evidence from US Labor Markets. *SSRN Electronic Journal*.
<https://doi.org/10.2139/ssrn.2940245>

Economic Complexity No. 2 Homework

Sarkar, S., Cottineau-Mugadza, C., & Wolf, L. J. (2024). Spatial inequalities and cities: A review. *Environment and Planning B: Urban Analytics and City Science*, 51(7), 1391-1407.

Research Question: The aim is to focus on the spatial-geographic dimension of how inequalities play out in urban space, and to identify and organise what it is about cities that fuels inequalities.

Data: 17 empirical papers, across 9 countries and several hundred cities.

- OpenStreetMap (OSM) – street networks & transit
- Mobile phone data – mobility & accessibility
- Census/microdata – demographics & income

Methods: Summary of methodologies used in the analyzed articles, compare them, and identify gaps.

Built their own framework to help distinguish between simple spatial differences and deeper forms of structural unfairness, pushing future research toward normative, justice-oriented analysis

Conceptual gradient: Disparity → Inequality → Injustice

Key Findings: Identify key factors that shape inequalities: (1) **Location**, (2) **Movement**, (3) **Infrastructure**

High-income groups enjoy better accessibility and shorter travel times.

Urban sprawl increases energy, transport, and health inequalities.

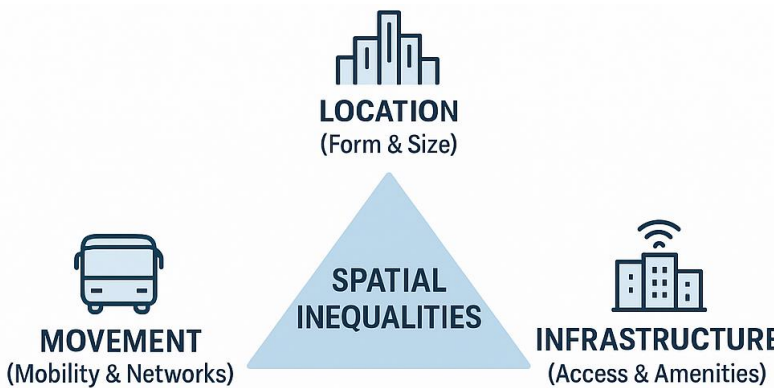
Transit-oriented and compact urban forms improve accessibility and sustainability.

Urban structure and transport networks can be powerful levers to reduce inequality.

Future urban science should: Integrate AI/ML responsibly,

Strengthen normative (justice-based) approaches,

Emphasize public participation and equitable policy design.



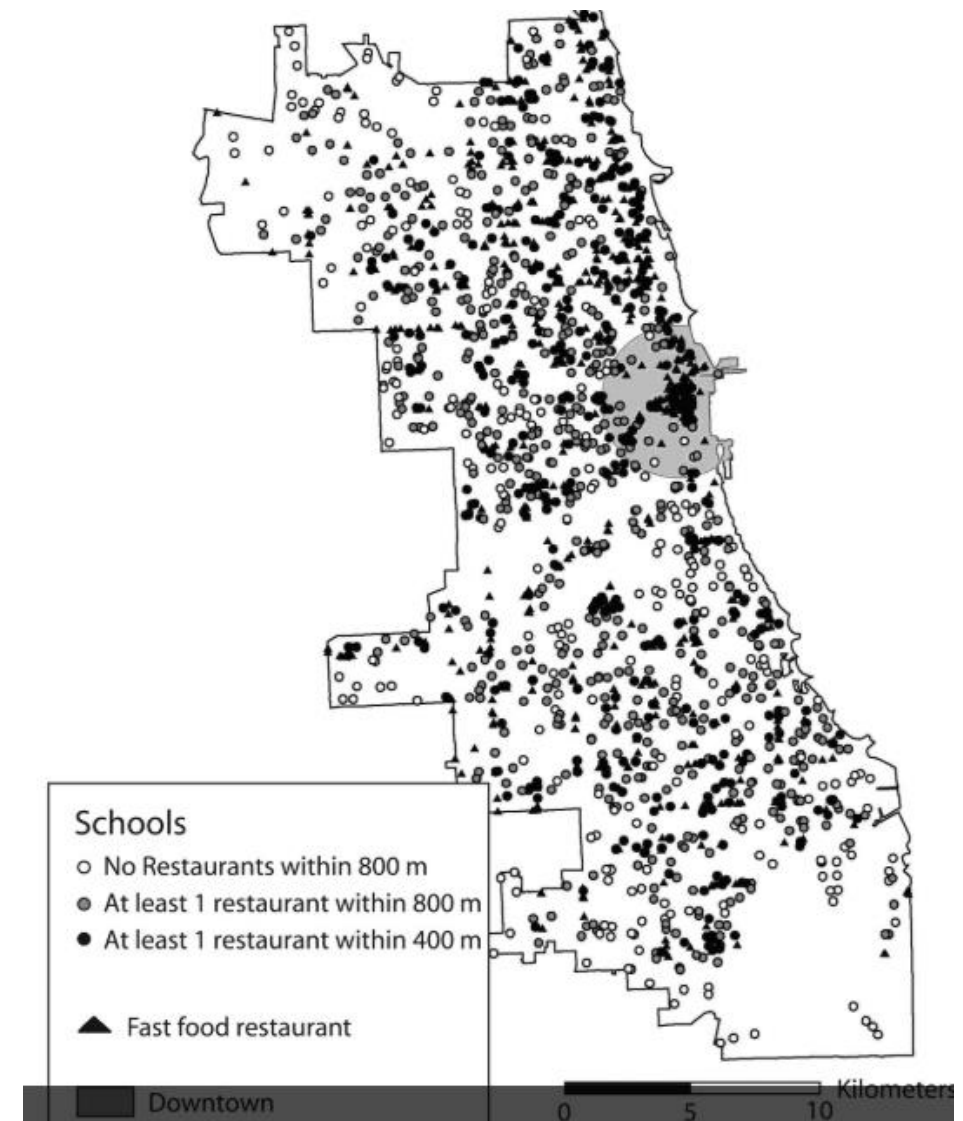
Disparity → Inequality → Injustice

**Austin, S. B., Melly, S. J., Sanchez, B. N., Patel, A., Buka, S.,
& Gortmaker, S. L. (2005). Clustering of fast-food
restaurants around schools: a novel application of spatial
statistics to the study of food environments.**

Second economic complexity homework

Nusser Csanád

- **Research question:** Do fast-food restaurants concentrate more around schools?
- **Data:** Geocoded 613 restaurants and 1292 schools in Chicago
- **Method:** The authors used the bivariate K function method to quantify the degree of clustering
- **Results:** A significantly greater number of fast-food restaurants are located within a short distance from schools than would be expected if there were no spatial dependence



Europe's Scaleup Geography and the Role of Access to Talent

Author: prof. Zoltán Cséfalvay

Source: J. Glückler, R. Panitz (eds.), Knowledge and Digital Technology,
Knowledge and Space 19, Ch. 6., pp. 107-130,
https://doi.org/10.1007/978-3-031-39101-9_6

Presenting: Ákos Virág

22. October 2025

Short summary: Cséfalvay (2024)

Research questions	1) What is the geography of European scaleups across cities (FUAs)?		
	2) How does access to locally available talent shape that landscape?		
Data	Dealroom.co snapshot (2021): 13,851 European scaleups mapped to 166 EU–OECD FUAs <i>Performance metrics:</i> number of scaleups, total funding, number valued over €200 m.		
Methods	K-means clustering	Talent proxies	Simple linear regression
Key Findings	Size matters	Location matters	Knowledge matters
	Pivotal factors develop first in a self-sustaining and then a self-reinforcing system.	Strong scaleup performance is embedded in Europe's economic and territorial disparities.	An influencing factor is the upstream stemming from the university students in the cities in question

Elaboration on key findings: Cséfalvay (2024)

Top hubs capture a **disproportionate share of outcomes** ($\approx 61\%$ of all scaleups, 71% of funding, 68% of €200m+ firms concentrated in ~ 15 cities).

Clear **West–East** and **North–South** gradient, so national capitals in Western/Northern Europe far outpace peers. cities plugged into strong cross-border networks benefit most; peripherals **depend on proximity and links** to nearby hubs..

“Upstream” talent and “downstream” venture outcomes results in cities that **attract, train, and retain students** generate larger founder pools and more scaleups.

Goal: Strengthening of university-startup bridges

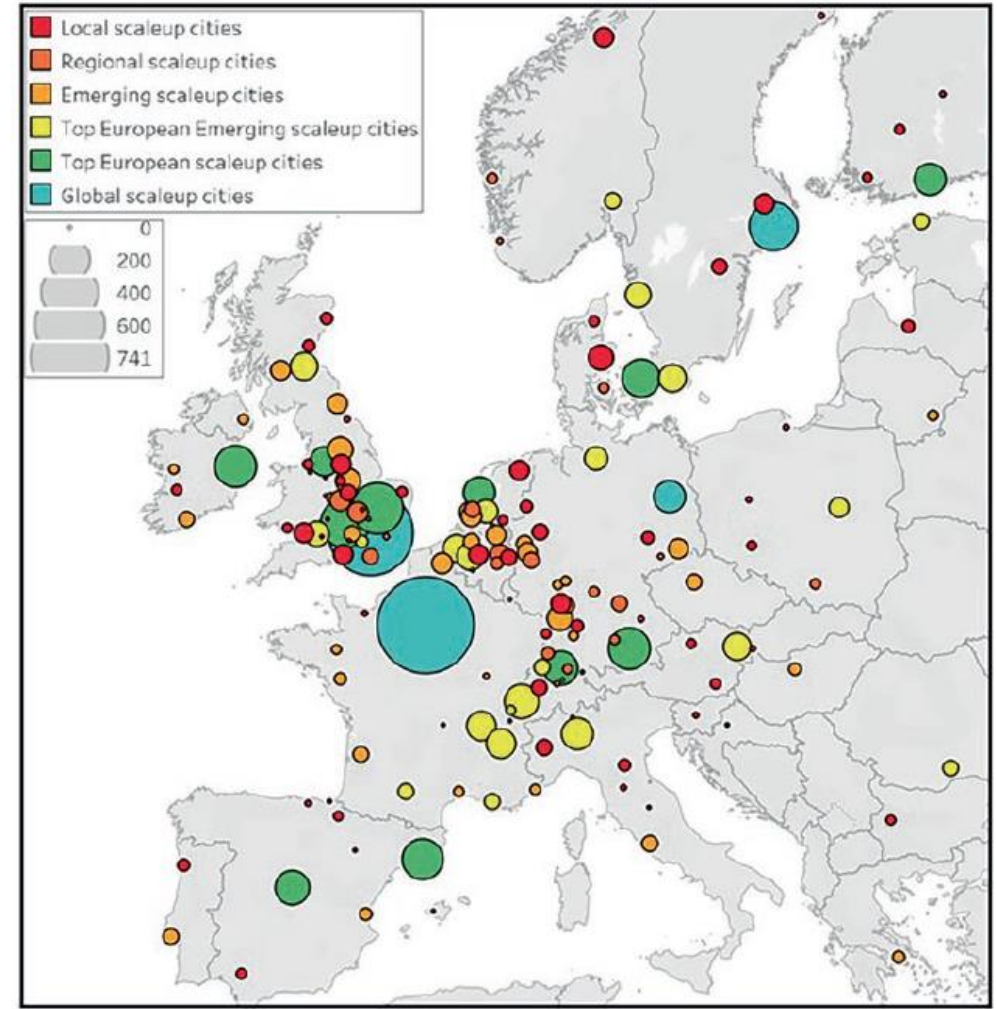


Fig. 6.9 Number of founders attended a university of the scaleup city who raised more than €10 million in funding, Europe, 2021. Source: Author's own calculation based on Dealroom data retrieved June 2, 2021 from <https://dealroom.co> and EU-OECD FUA classification. Design by author

Patterns of Spatial Concentration and Drivers of China's Migrant Population: Evidence from the Greater Bay Area Hinterland

AUTHORS: CHAO GE & FAN YANG (2025)

BY: MÁTYÁS BALOGH

Introduction and Research Question

- ❑ Rapid Urbanization and Migration in China

- ❑ China's urbanization rate rose from **<20% (1980s) to >60% today**
- ❑ Migration is a **key urban phenomenon**: 26% of China's population are „migrants“ (≈376 million)
- ❑ **Greater Bay Area (GBA)** = one of China's most dynamic industrial and urban regions
- ❑ Focus city: **Dongguan** known as “the world factory”

- ❑ Paraphrased Research Question

What are the spatial patterns and driving factors behind the concentration of China's migrant population in Dongguan, the hinterland of the Greater Bay Area?

- ❑ Objectives:

- ❑ Identify **spatial clustering patterns** of migrant populations
- ❑ Analyze **economic, social, and geographic drivers** of migration
- ❑ Explore how **local governance and industrial structure** shape spatial concentration

Data

- ❑ Multi-source Data (2021, Dongguan City) - sample size: 560 communities + admin. villages
 - ❑ **Official statistics:** Dongguan Subdistrict & Township Yearbooks (2021)
 - ❑ **Socioeconomic data:** industrial enterprises, assets, incomes
 - ❑ **Spatial data:** satellite imagery, OpenStreetMap boundaries
 - ❑ **Housing & land prices:** Baidu Maps + Anjuke (Python scraping)
- ❑ Key variables (10 factors)
 - ❑ *Industry scale:* total numbers of enterprises, industrial enterprises
 - ❑ *Economic level:* total assets, governance income, housing prices
 - ❑ *Regional advantage:* distance & travel time to Shenzhen
 - ❑ *Land use:* industrial land ratio, land-use diversity



Methodology

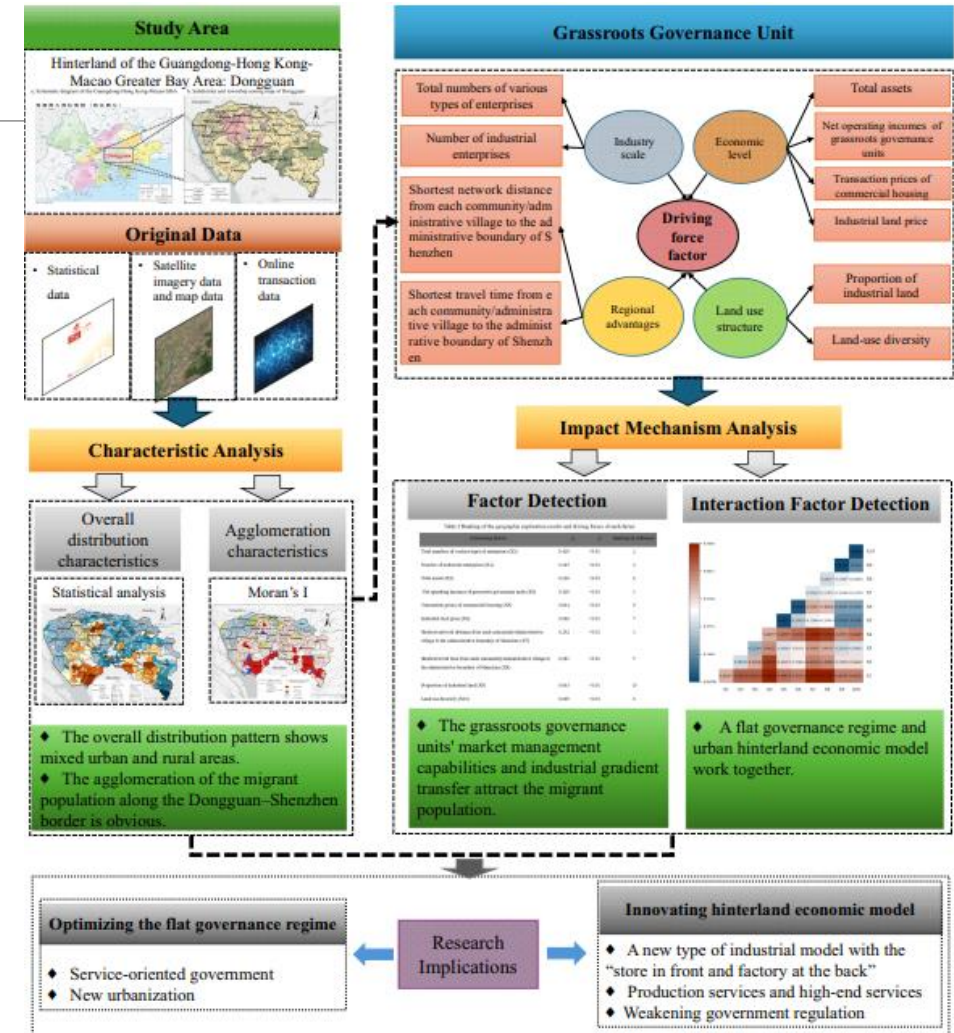
Methods:

- ❑ **Spatial autocorrelation (Moran's I)** -> Measures population clustering and spatial dependence
- ❑ **Geographic detector model** -> Quantifies explanatory power of each factor (q-values)
- ❑ **Interaction analysis** -> Tests combined effects (e.g. enterprises × governance income)

Equation example (Geographic detector):

$$q = 1 - \frac{\sum_{h=1}^L N_h \sigma_h^2}{N \sigma^2}$$

(higher q = stronger influence on spatial concentration)



Key Findings

- ❑ Migrant population **clustered near Shenzhen border** and city centre
- ❑ *Global Moran's $I = 0.37$ ($p < 0.01$)* -> significant agglomeration
- ❑ Top drivers (q-values):
 - ❑ Net operating income of governance units(0.43)
 - ❑ Number of enterprises(0.420)
 - ❑ Proximity to Shenzhen(0.292)
- ❑ Local governance units act both as **service providers and economic actors**
- ❑ Industrial spillovers from Shenzhen support the “*store in front, factory at back*” model

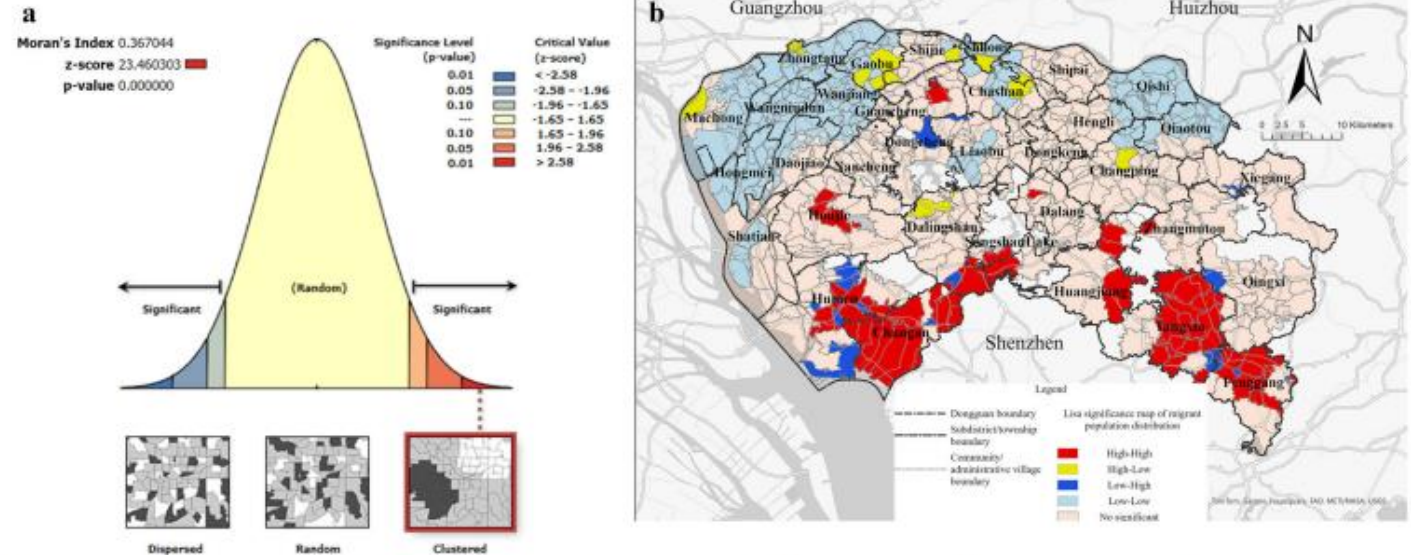


Fig. 4 Agglomeration characteristics of Dongguan's community/administrative village migrant population. **a** Global Moran's I result. **b** Local Moran's I result.

Implications

- ❑ Grassroots governance shapes **inclusive and flexible urbanization**
- ❑ Migrant concentration highlights the need for:
 - ❑ Balanced **public service supply**
 - ❑ **Sustainable development** of urban hinterlands
- ❑ **Urban complexity:** Interaction of people, land, and industry forms a self-reinforcing urban system

-> Dongguan represents a new type of urbanization driven by economic complexity and governance innovation

Source

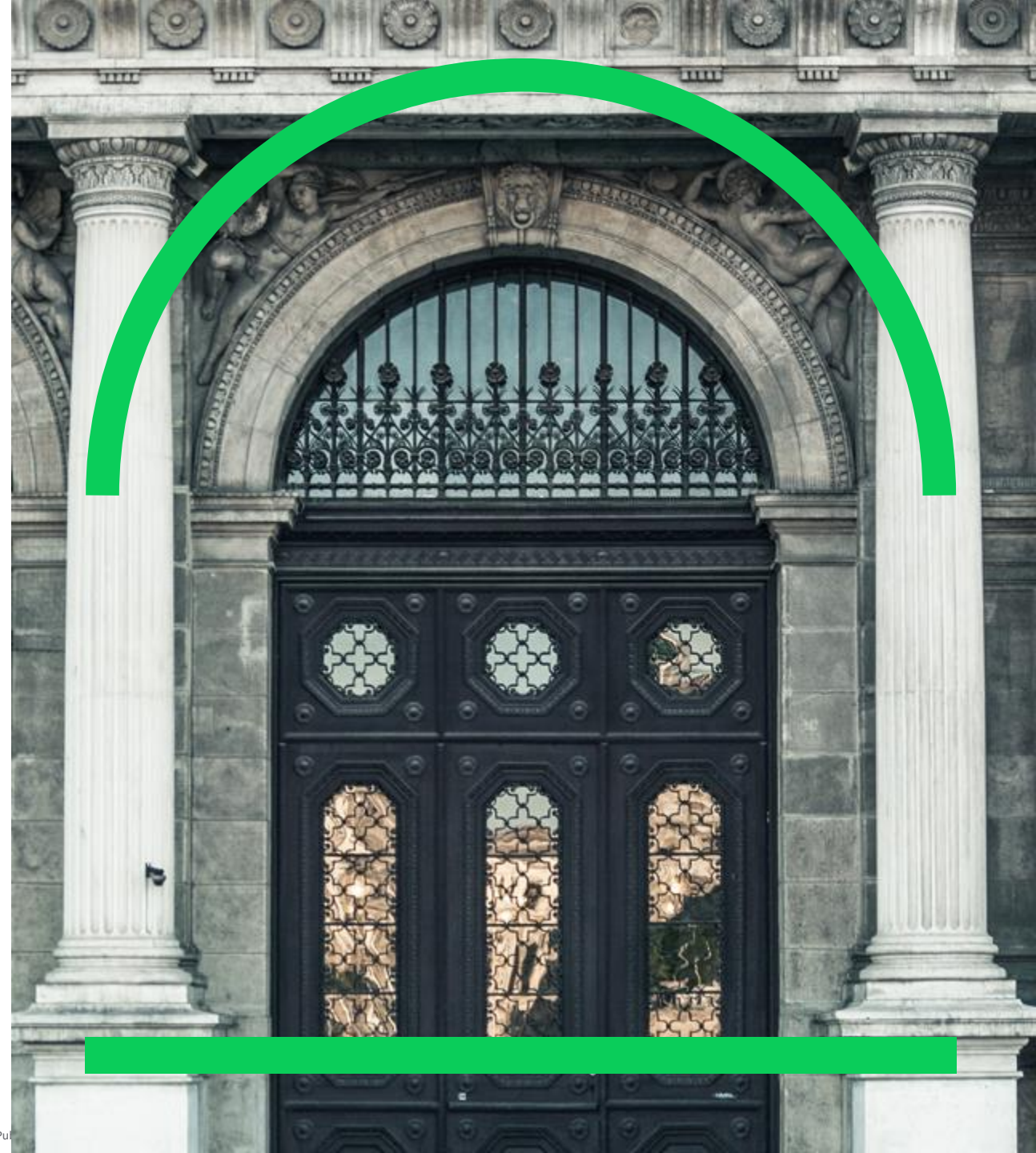
Ge, C., & Yang, F. (2025). Patterns of spatial concentration and drivers of China's migrant population: evidence from the Greater Bay Area hinterland. *Humanities and Social Sciences Communications*, 12(1). <https://doi.org/10.1057/s41599-025-04702-5>

Work of the Past, Work of the Future

by David H. Autor (2019)

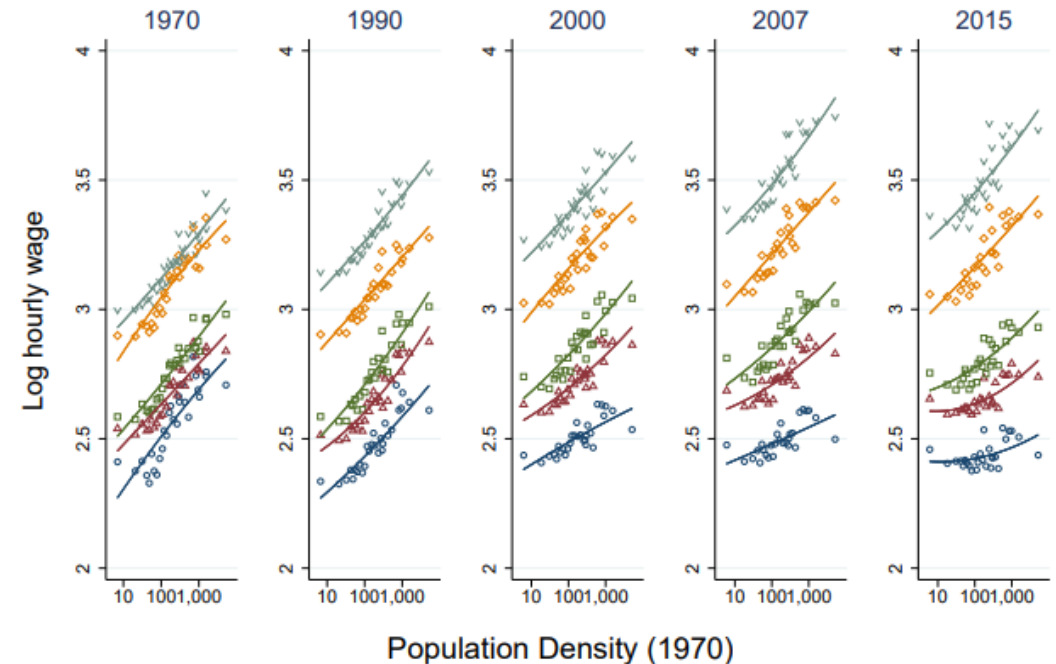
Levente Szabó

19th of October 2025



Summary

- **Research question:** How has technological change, especially automation, transformed the structure and geography of work in the United States?
- **Data:**
 - U.S. Census and American Community Survey (ACS) microdata (1980-2015)
 - Bureau of Labor Statistics (BLS) occupational employment data
 - O*NET task measures → classify jobs by routine, manual, or abstract task intensity.
 - Covers men and women aged 16-64 across U.S. metropolitan areas.
- **Methodology:**
 - Task-based decomposition: Uses O*NET data to classify occupations by routine vs. non-routine tasks and track employment and wage changes (1980-2015) using U.S. Census and ACS data.
 - Spatial comparison: Examines how automation-driven job polarization differs across city sizes.
- **Results:**
 - Middle-skill routine jobs declined sharply, while high- and low-skill jobs expanded → strong job polarization.
 - Large urban areas captured most high-wage, high-skill growth, leaving smaller regions behind.
 - Automation and digitalization are key forces driving this new geography of work and rising inequality.



R. Martin & P. Sunley (2023):

Capitalism Divided? London, financialisation and the UK's spatially unbalanced economy

<https://www.tandfonline.com/doi/epdf/10.1080/21582041.2023.2217655?needAccess=true>

1. Research question:

- Has London's economic success benefited the regions?
- Trickle down effect or „finance curse“?
- Has the government's „Levelling up“ policy been successful?

2. Data:

- Historical time series data
- Trade balance statistics
- Regional per capita Gross-value added (GVA)

3. Methodology:

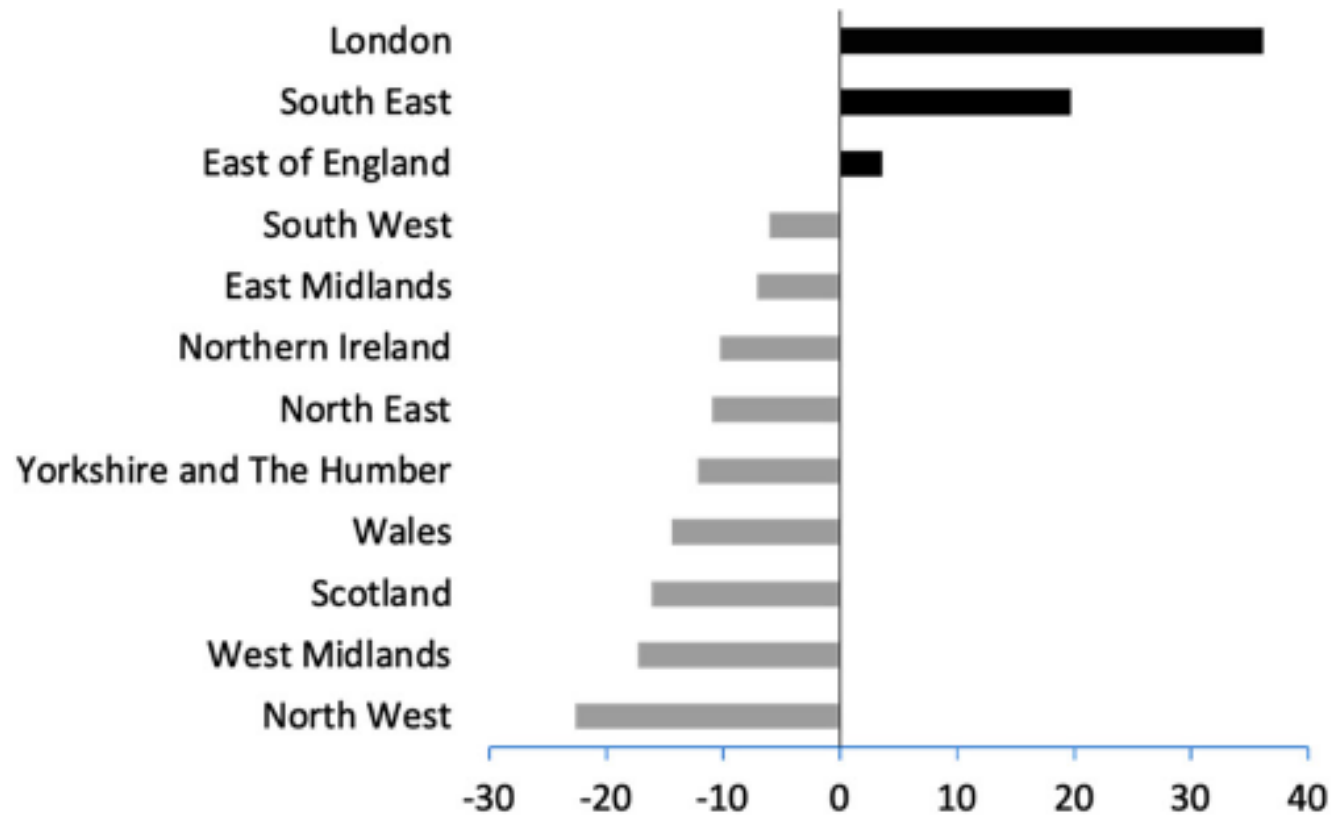
- Historical trend analysis ← time series reaching back as far as the 1870s
- Literature analysis ← integrating other author's finding regarding the „finance curse“
- Policy critique

4. Findings:

- London's lead has been increasing since the 80s (25% of the UK's GDP and 38% of Englands GDP)
- Misallocation of capital (London receives highest per capita amount of gov. spending)
- Insufficient policies, deeply institutionalised problems
- Gravitational pull

„At the same time, regional and sub-regional disparities in economic prosperity in the UK are amongst the largest in the OECD nations, far greater than in other major advanced economies such as Germany, France, Italy, Japan and the USA, and only exceeded by those in Hungary, Turkey and Colombia (OECD, 2020).“

Net Fiscal Balance (£billion), FYE 2020



Net Public Fiscal Surplus (Tax Revenue minus Expenditure): London and the Other UK Regions, Financial Year 2019/2020

Smart Specialisation Strategies and regional knowledge spaces: how to bridge vision and reality

Keungoui Kim, Chiara Ferrante & Dieter F. Kogler

Economic Complexity
Homework 3

Budapest, 2025.10.20.

Nagy Zalán Zsolt
ILCRVD

Summary

The study investigates the extent to which **Smart Specialisation Strategies** (S3, EU policy for regional innovation) **align with the actual technological and other relevant capacities of European regions.**

Data:

- 164 European regions, scrutinising the period from 2011-2015.
- European Patent Office PATSTAT database – **technological knowledge production via patents**
- European Commission S3 platform – **S3 priority sectors**

Methodology: evolutionary economic geography and network analysis

- Framework 1: **Centrality Analysis** – tests whether S3 policies focus on central technologies of the local innovation network.
- Framework 2: **Specialisation and potential analysis** – assesses whether S3 priorities focus on existing strengths or potential diversification opportunities.

The study found that **many S3 strategies do not align with the regional capabilities. Innovation remains spatially concentrated** in regions with dense and diverse knowledge places, with **institutional inequality** having a strong influence on that persistence.

Key figures

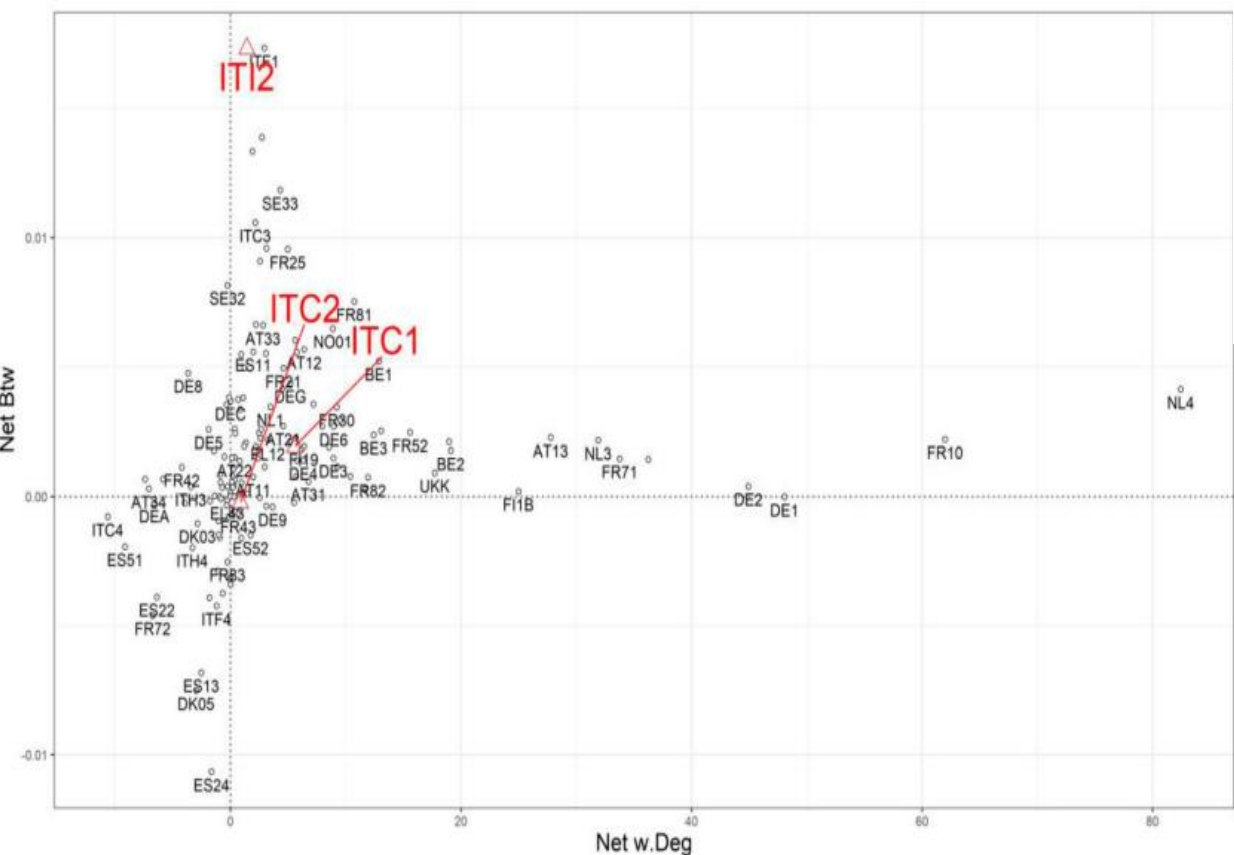


Figure 1. Net weighted degree centrality and net betweenness centrality of Smart Specialisation and knowledge space

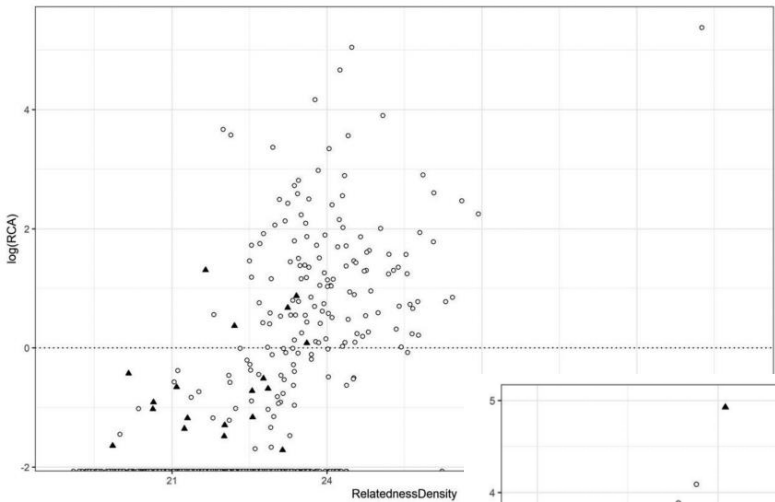


Figure 2. RCA and relatedness density of Umbria

Figure 3. RCA and relatedness density of Valle d'Aosta

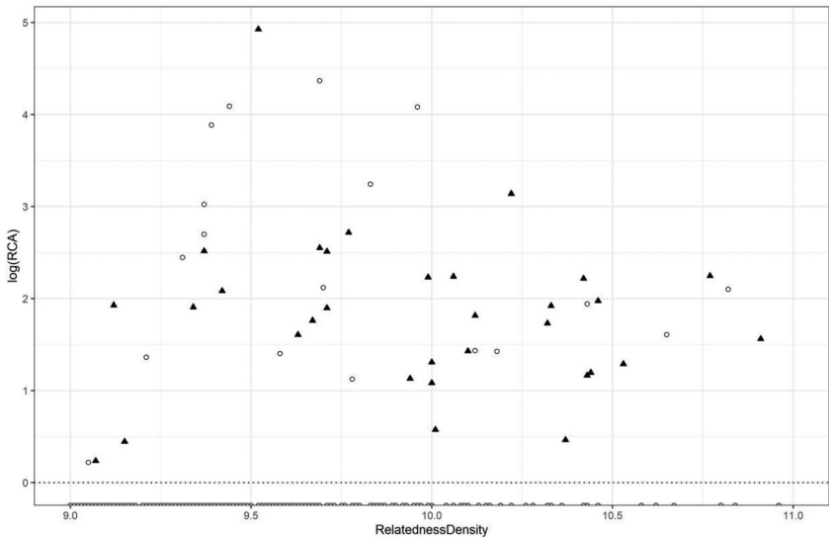
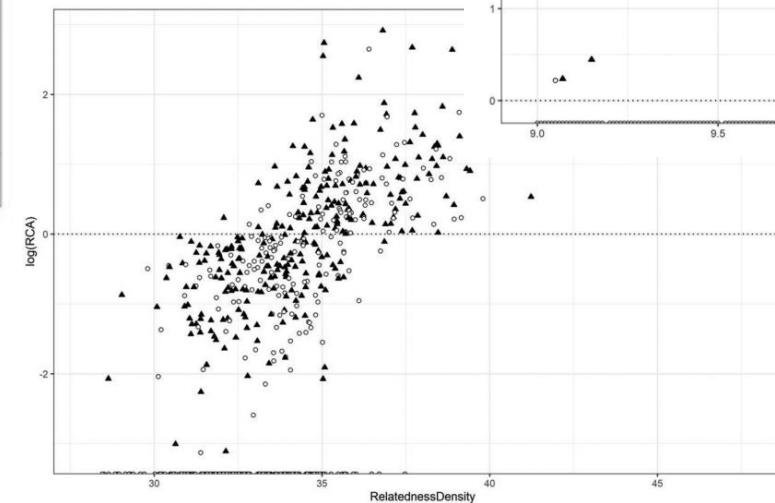


Figure 4. RCA and relatedness density of Piemonte

Thank you for your attention!

Source:

Keungoui Kim, Chiara Ferrante & Dieter F. Kogler (2024) Smart Specialisation Strategies and regional knowledge spaces: how to bridge vision and reality, *Regional Studies*, 58:12, 2501-2517, DOI: 10.1080/00343404.2024.2355985

Growth, Innovation, and the Pace of Life in Cities

Bettencourt et al. EC2025 HW2

Research question

- Do cities obey universal scaling laws with population size?
- Hypothesis: many urban indicators Y scale as $Y = Y_0 \cdot N^\beta$ across countries and time.
- Interpretation: β clusters into classes reflecting mechanisms — sublinear (infrastructure efficiencies), linear (individual needs), superlinear (social interaction, innovation).

Data

- Urban systems: U.S., European Union, China, Germany (various years, 1990–2003).
- Indicators: wages, GDP, bank deposits, patents, R&D, crime, AIDS cases; infrastructure: cables, road surface; household use: electricity, water, housing.
- Cities defined as integrated labor markets (e.g., MSAs/LUZs). Ordinary least squares on log–log data with heteroskedasticity correction.

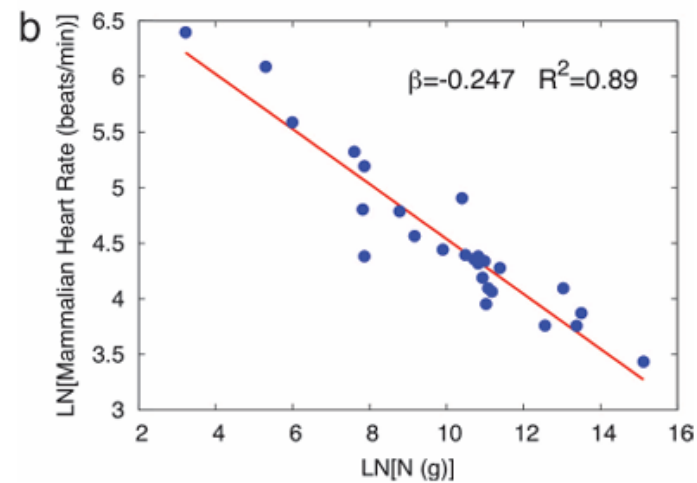
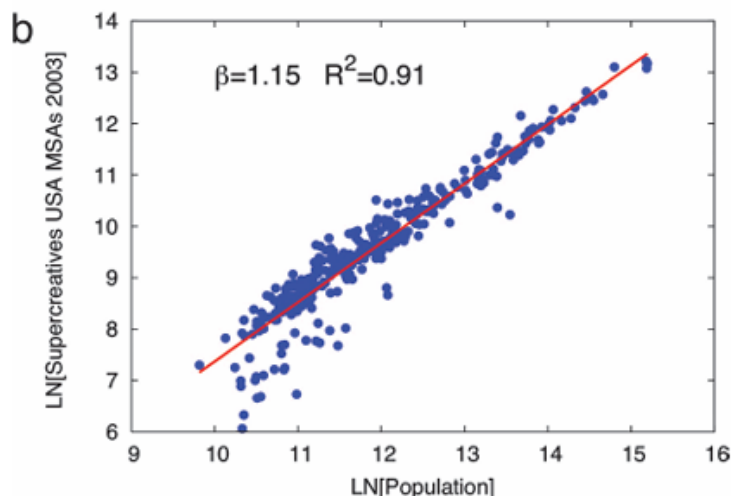
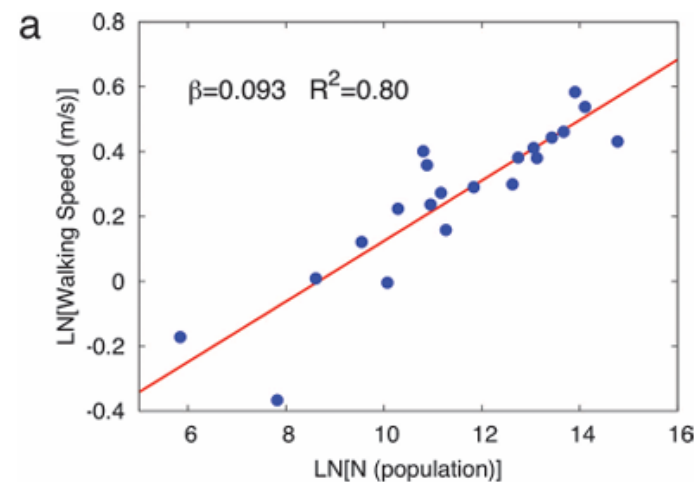
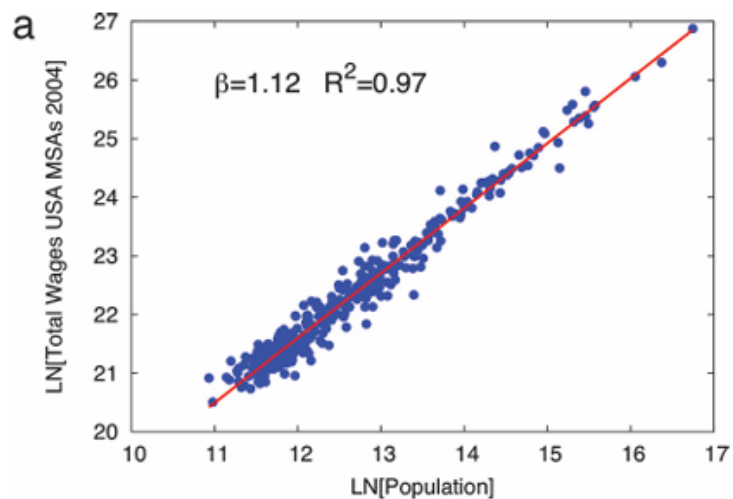
Scaling results

Table 1. Scaling exponents for urban indicators vs. city size

Y	β	95% CI	Adj- R^2	Observations	Country-year
New patents	1.27	[1.25,1.29]	0.72	331	U.S. 2001
Inventors	1.25	[1.22,1.27]	0.76	331	U.S. 2001
Private R&D employment	1.34	[1.29,1.39]	0.92	266	U.S. 2002
"Supercreative" employment	1.15	[1.11,1.18]	0.89	287	U.S. 2003
R&D establishments	1.19	[1.14,1.22]	0.77	287	U.S. 1997
R&D employment	1.26	[1.18,1.43]	0.93	295	China 2002
Total wages	1.12	[1.09,1.13]	0.96	361	U.S. 2002
Total bank deposits	1.08	[1.03,1.11]	0.91	267	U.S. 1996
GDP	1.15	[1.06,1.23]	0.96	295	China 2002
GDP	1.26	[1.09,1.46]	0.64	196	EU 1999–2003
GDP	1.13	[1.03,1.23]	0.94	37	Germany 2003
Total electrical consumption	1.07	[1.03,1.11]	0.88	392	Germany 2002
New AIDS cases	1.23	[1.18,1.29]	0.76	93	U.S. 2002–2003
Serious crimes	1.16	[1.11, 1.18]	0.89	287	U.S. 2003
Total housing	1.00	[0.99,1.01]	0.99	316	U.S. 1990
Total employment	1.01	[0.99,1.02]	0.98	331	U.S. 2001
Household electrical consumption	1.00	[0.94,1.06]	0.88	377	Germany 2002
Household electrical consumption	1.05	[0.89,1.22]	0.91	295	China 2002
Household water consumption	1.01	[0.89,1.11]	0.96	295	China 2002
Gasoline stations	0.77	[0.74,0.81]	0.93	318	U.S. 2001
Gasoline sales	0.79	[0.73,0.80]	0.94	318	U.S. 2001
Length of electrical cables	0.87	[0.82,0.92]	0.75	380	Germany 2002
Road surface	0.83	[0.74,0.92]	0.87	29	Germany 2002

Data sources are shown in [SI Text](#). CI, confidence interval; Adj- R^2 , adjusted R^2 ; GDP, gross domestic product.

Output & pace of life scale with city size



Growth regimes implied by scaling (β)

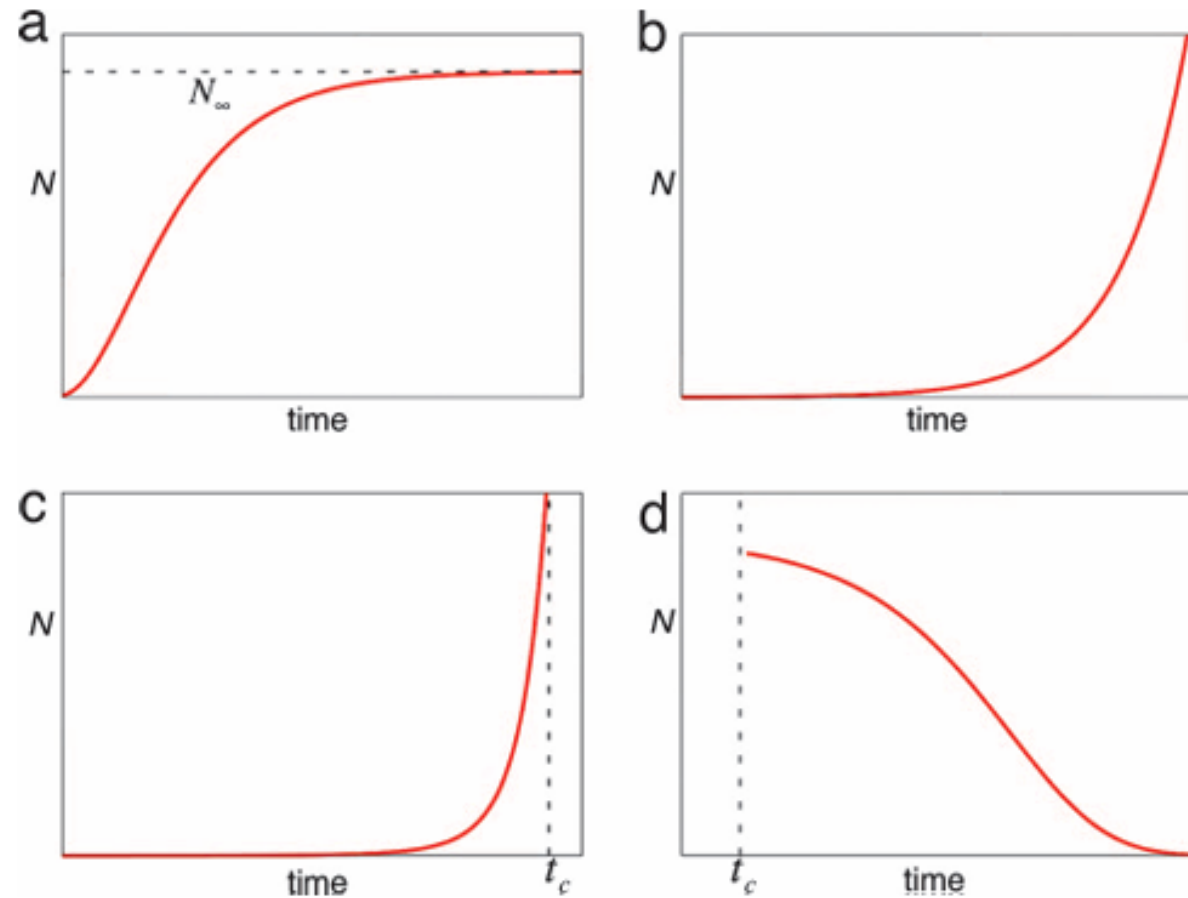
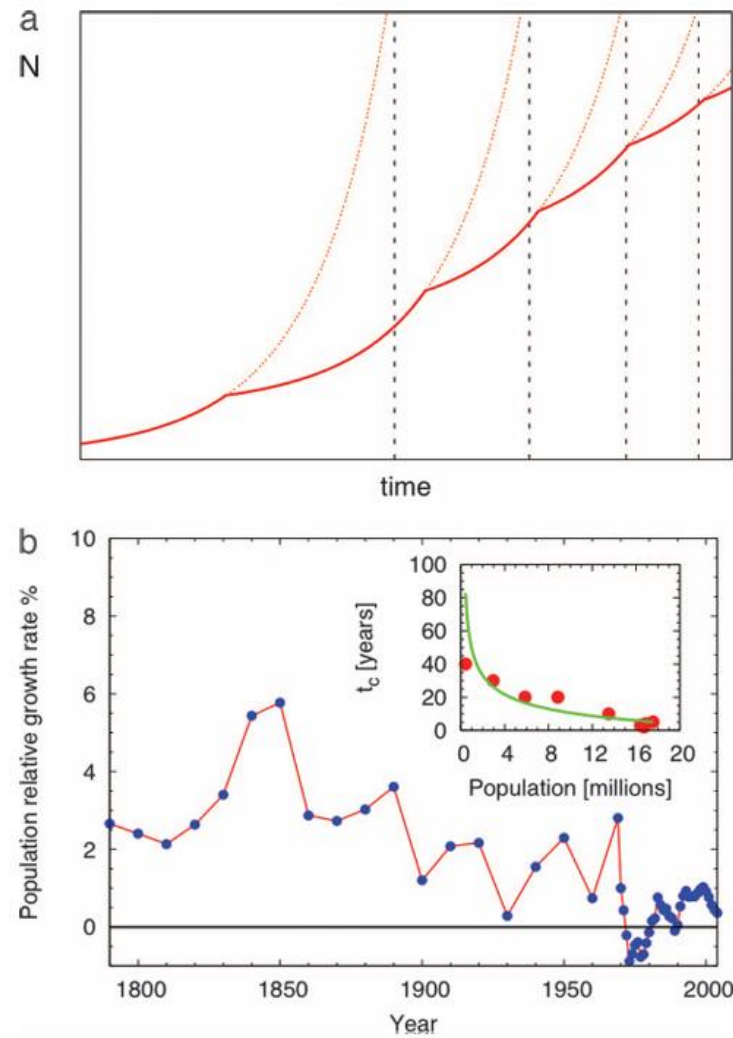


Fig. 3. Regimes of urban growth. Plots of size N vs. time t . (a) Growth driven by sublinear scaling eventually converges to the carrying capacity N_∞ . (b) Growth driven by linear scaling is exponential. (c) Growth driven by superlinear scaling diverges within a finite time t_c (dashed vertical line). (d) Collapse characterizes superlinear dynamics when resources are scarce.

Innovation cycles to sustain superlinear cities



Key findings

- Superlinear ($\beta \approx 1.1\text{--}1.3$): innovation/wealth and social 'bads' rise faster than population (wages, patents, crime).
- Linear ($\beta \approx 1$): individual needs (employment, housing, household utilities).
- Sublinear ($\beta \approx 0.8$): infrastructure economies of scale (roads, cables, gas stations).
- Pace of life increases with size (e.g., walking speed $\beta \approx 0.09$).
- Policy: larger cities are not just 'bigger'—they are different per capita; targets should adjust for expected scaling baselines.

Citation

- Bettencourt, L. M. A., Lobo, J., Helbing, D., Kühnert, C., & West, G. B. (2007). Growth, innovation, scaling, and the pace of life in cities. PNAS, 104(17), 7301–7306.



THE EFFECT OF HOME- SHARING ON HOUSE PRICES AND RENTS: EVIDENCE FROM AIRBNB

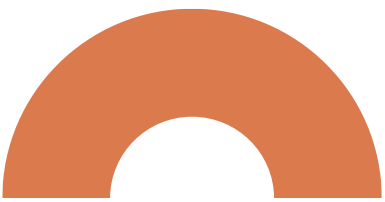
Ismet Ogulcan Togan



INTRODUCTION



Airbnb changed how housing is used in cities, letting owners switch homes from long-term tenants to short-term visitors. This paper asks whether that shift pushes up rents and house prices, and tests the idea using ZIP-level Airbnb data linked to Zillow rents/prices and an IV strategy (Google Trends × touristiness) to isolate causal effects. The results show small but meaningful increases in both rents and prices—strongest in areas with lower owner-occupancy, consistent with a reallocation of housing supply to tourism.



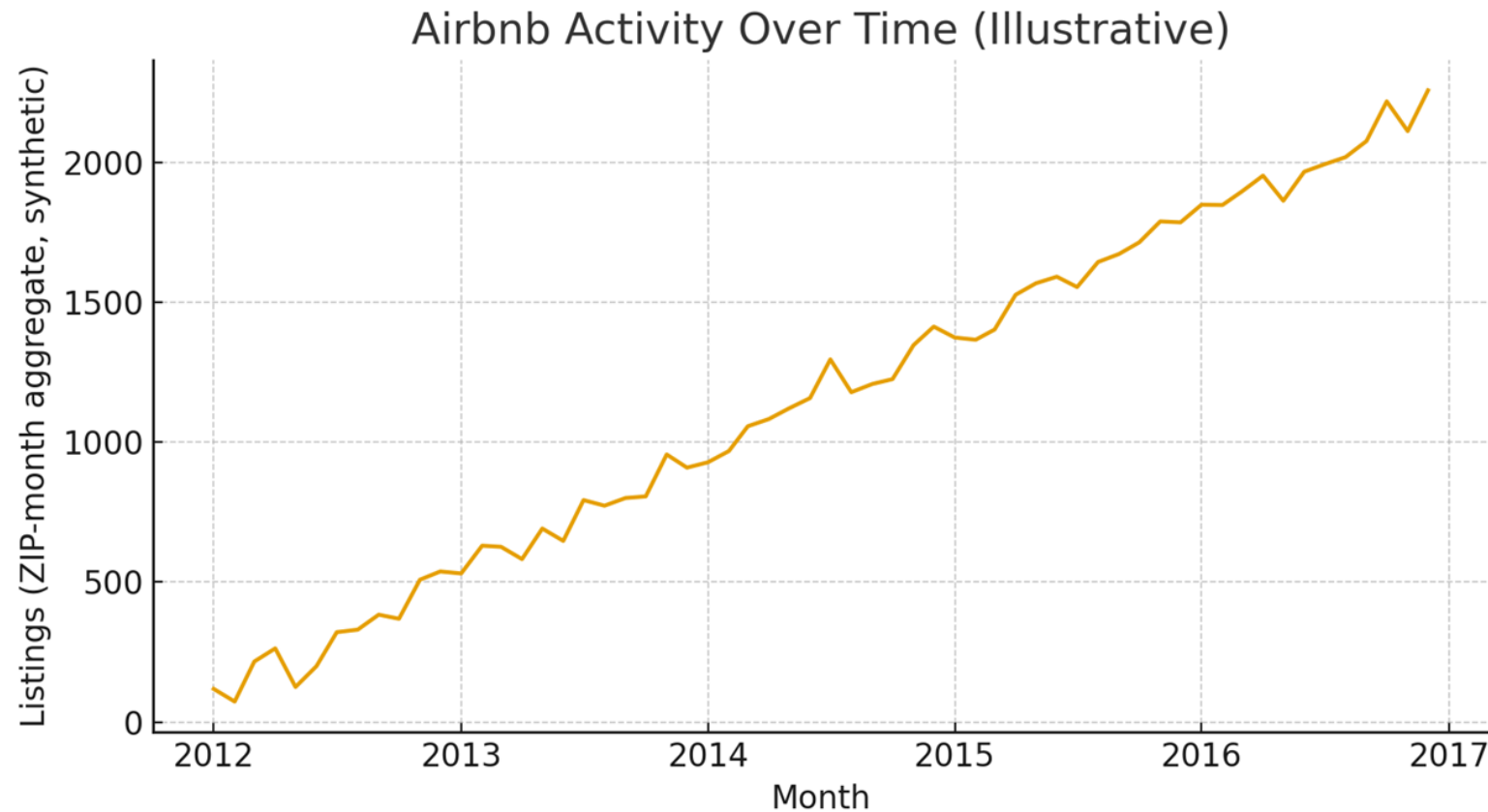
METHODOLOGY

- Panel regressions with ZIP fixed effects and CBSA×year×month fixed effects to absorb time-invariant neighborhood traits and metro-wide shocks.
- 2SLS/IV strategy: (Google Trends “airbnb” × touristiness) predicts plausibly exogenous shifts in local Airbnb supply; coefficients are allowed to vary with owner-occupancy via interaction terms to test the hypothesized reallocation channel.
- Robustness: Alternative time windows, different city sizes, sub-samples (central vs. peripheral ZIPs), alternative Airbnb measures, and controls for other platforms (e.g., VRBO/HomeAway). They also probe lead/lag patterns and demographic trends to address gentrification/anticipation concerns.

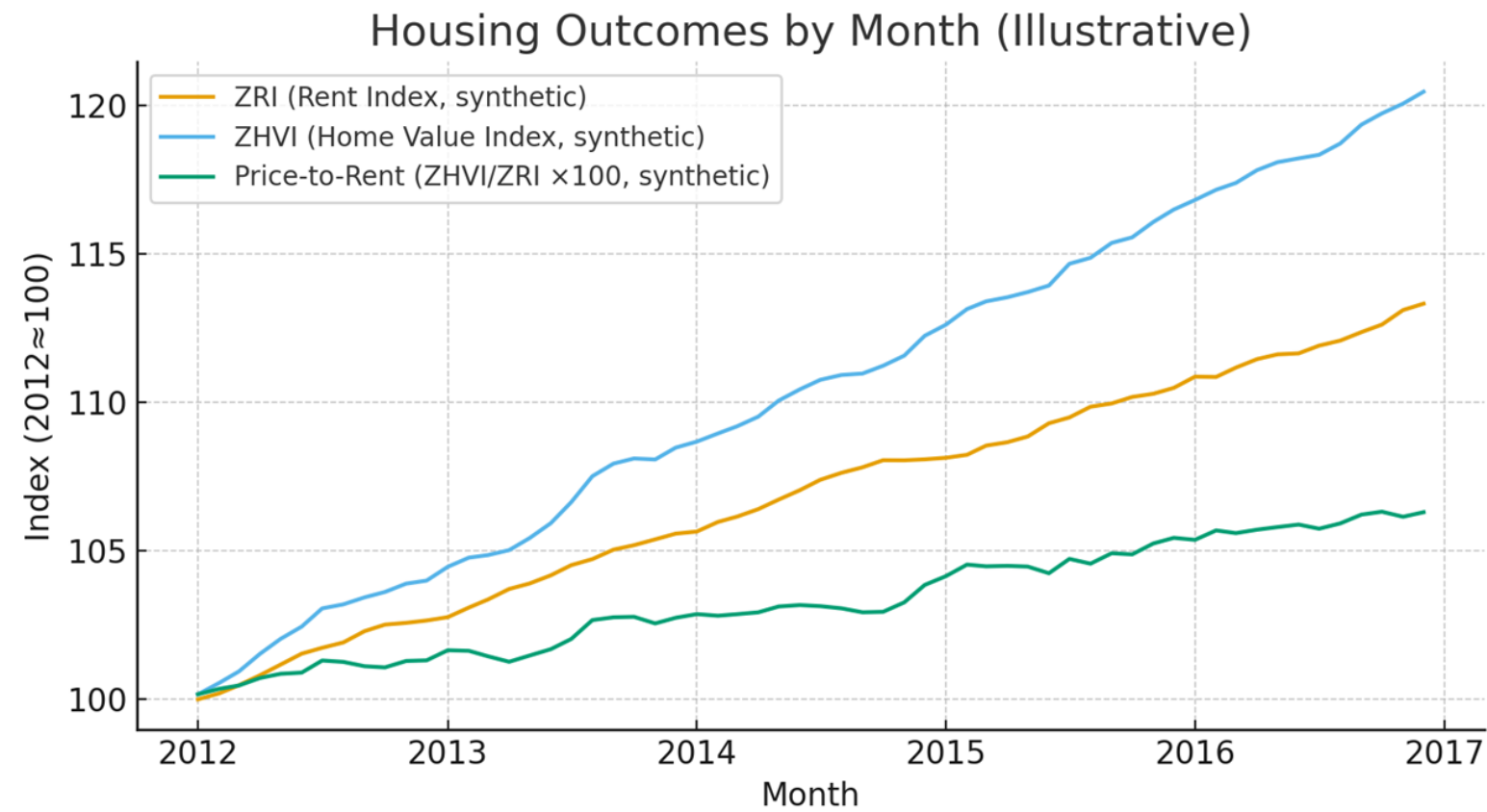
DATA

- Airbnb activity: Monthly counts of listings at the ZIP-code level across major U.S. metros, focusing on 2012–2016 when the platform scaled rapidly.
- Housing outcomes: Zillow’s ZIP-level Rent Index (ZRI) and Home Value Index (ZHVI); the paper also tracks the price-to-rent ratio (ZHVI/ZRI) to see whether prices respond more than rents.
- Instrument & controls: An instrument built from Google Trends searches for “airbnb” (time variation) interacted with each ZIP’s baseline touristiness (measured by NAICS 72 establishments around 2010).
Rich controls include ACS demographics (income, population, education, employment), owner-occupancy and vacancy types (seasonal vs. for-rent), plus tourism demand proxies (hotel occupancy, airport traffic, TripAdvisor review volumes). These help separate Airbnb-driven effects from general tourism or neighborhood change.

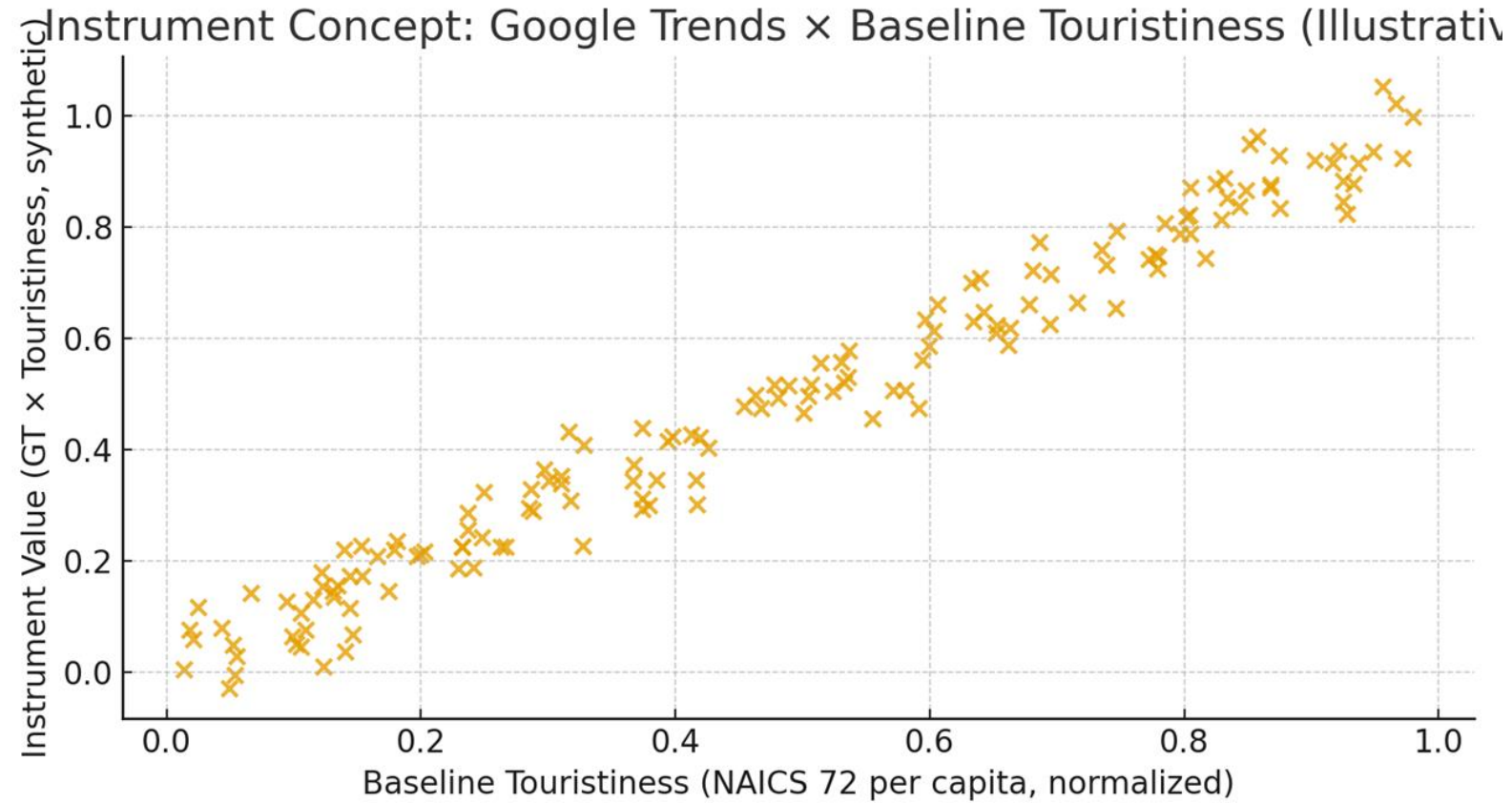
Data Visual 1 — Airbnb Activity Over Time (Illustrative)



Data Visual 2 — ZRI, ZHVI, and Price-to-Rent (Illustrative)



Data Visual 2 — ZRI, ZHVI, and Price-to-Rent (Illustrative)



- Robustness & interpretation. Results remain when adding tourism controls, using alternate Airbnb measures, slicing the sample, and accounting for other platforms. The moderation by owner-occupancy strongly supports the reallocation story (investor-heavy areas are more affected).

- Positive effects on rents and prices. At median owner-occupancy ($\approx 72\%$), a 1% increase in Airbnb listings is associated with +0.018% in rents and +0.026% in prices. Where owner-occupancy is lower ($\approx 56\%$), effects are larger (+0.024% rents; +0.037% prices); where higher ($\approx 82\%$), effects are smaller (+0.014%; +0.019%).

MAIN FINDINGS

- Mechanism—reallocation of supply. Airbnb growth coincides with more seasonal/occasional vacancies, fewer units available for long-term rent, and no meaningful change in total housing stock, consistent with owners shifting use from long-term tenants to short-term visitors.

- Prices rise a bit more than rents. The price-to-rent ratio increases by about 0.01% per 1% rise in listings, suggesting higher option value of housing when short-term use is profitable. Economic magnitude. Given typical annual growth in listings ($\sim 28\%$ in the period), Airbnb expansion can account for roughly 0.5% of annual rent growth and 0.7% of annual price growth for the median ZIP—modest in macro terms but meaningful in tight neighborhoods.



Thank You

For your attention

